



UNIVERSITÀ DI PISA

Laboratory of Data Science

Project Report: Decision Support Systems for Music Streaming

Group ID: 05

Module II

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Introduction & Architecture

The digital music streaming industry produces large volumes of heterogeneous and continuously evolving data, ranging from raw transactional streams to rich contextual metadata about artists, content, and user engagement. Extracting strategic value from such data requires a structured analytical architecture that supports data quality, scalability, and multidimensional analysis.

This project presents the end-to-end design and implementation of a Decision Support System (DSS) for music streaming analytics. The system enables the analysis of artist performance, song characteristics, temporal trends, and geographic consumption patterns through a unified data pipeline.

The architecture is organized into three tightly integrated layers. First, a **Data Preparation** layer implemented in Python performs data cleaning, enrichment, feature derivation, and profiling, transforming raw XML and JSON sources into analysis-ready datasets. Second, an **ETL layer** implemented using SQL Server Integration Services (SSIS) loads the data into a structured Data Warehouse while enforcing referential integrity and enabling complex transformations and statistical benchmarking. Third, a **Multidimensional Analysis layer** implemented with SQL Server Analysis Services (SSAS) supports advanced OLAP queries for growth analysis, portfolio evaluation, and seasonal comparisons.

Finally, the analytical results are exposed through interactive **Power BI dashboards**, enabling business users to explore trends in content performance, regional dynamics, lyrical complexity, and category evolution over time. Together, these components form a complete decision-support architecture that transforms raw operational data into actionable business insights.

Part I: Data Preparation (Python)

1 Assignment 1: Data Understanding

1.1 Objective

The objective of this assignment is to understand the structure and quality of the provided datasets (`artists.xml` and `tracks.json`) by:

- identifying missing values and their patterns,
- assessing whether missing data can be recovered or reasonably filled, and
- evaluating opportunities for external data integration (e.g., spatial encodings such as Uber H3) and feature derivation from existing attributes (e.g., lyrics-based metrics).

1.2 Data Sources and Structure

Two datasets were analyzed:

- **Artists dataset** (`artists.xml`) containing artist-level metadata such as identity, demographics, activity period, and geographic information (birthplace and coordinates).
- **Tracks dataset** (`tracks.json`) containing track-level metadata (titles, albums, release dates), lyrical fields and derived text metrics, and melodic/audio descriptors (e.g., BPM, loudness, spectral features).

The analysis was conducted in Python using XML parsing for the artists dataset and JSON loading for the tracks dataset.

1.3 Missing Values Assessment

To assess dataset completeness, all records were scanned and missing values were counted per field. Values were treated as missing if they were `None`, empty strings, or common placeholders (`null`, `NULL`, `\N`). Listing 1 shows the procedure used to detect missing values.

Listing 1: Missing Value Detection

```
1 missing_counts = {}
2 for track in tracks_data:
3     for key, value in track.items():
4         if value in [None, '', 'null', 'NULL', '\\N']:
5             if key not in missing_counts:
6                 missing_counts[key] = 0
7                 missing_counts[key] += 1
```

Missing values by field are summarized in Table 1 for the artists dataset and Table 2 for the tracks dataset.

Table 1: Missing Values per Field: Artists Dataset (artists.xml)

Field	Missing Count
id_author	0
name	0
gender	0
birth_date	31
birth_place	32
nationality	33
description	18
active_start	54
active_end	104
province	34
region	36
country	34
latitude	32
longitude	32

1.4 Interpretation of Missingness and Expected Impact

1.4.1 Artists Dataset

Missingness is concentrated in:

- biographical fields (e.g., `birth_date`, `description`, activity dates), and
- geographic fields (`birth_place`, `province`, `region`, `country`, `latitude`, `longitude`).

This pattern indicates that some artists have partial profiles where basic identifiers are present, but supporting descriptive and geographic information is incomplete.

1.4.2 Tracks Dataset

The tracks dataset exhibits substantial missingness in:

- `featured_artists`,
- temporal components (`year`, `month`, `day`), and
- several audio features (e.g., BPM, rolloff, loudness) for a subset of tracks.

A small number of tracks also lack lyrics, which likely corresponds to instrumental tracks.

1.5 Recoverability and Filling Feasibility

1.5.1 Artists: Geographic Recovery and Spatial Integration

Several missing fields in the artists dataset are realistically recoverable:

- If `birth_place` is available, missing coordinates and administrative fields can be recovered via geocoding.
- If coordinates are present but textual location fields are missing, reverse geocoding can infer country, region, and province.

Table 2: Missing Values per Field: Tracks Dataset (tracks.json)

Field	Missing Count
id	0
id_artist	0
title	0
featured_artists	7649
primary_artist	0
language	105
album	1514
swear_IT	0
swear_EN	0
swear_IT_words	0
swear_EN_words	0
year	444
month	1197
day	1323
n_sentences	76
n_tokens	76
char_per_token	76
avg_token_per_clause	76
bpm	64
rolloff	64
flux	64
rms	64
flatness	64
spectral_complexity	64
pitch	64
loudness	64
album_name	78
album_release_date	78
album_type	78
disc_number	78
track_number	78
duration_ms	78
explicit	78
popularity	78
id_album	78
lyrics	3
streams@1month	0

- Once coordinates are available, the dataset becomes suitable for spatial enrichment using hierarchical spatial indexes such as Uber H3, enabling region-level grouping and aggregation.

1.5.2 Tracks: Metadata Recovery and Derived Features

For the tracks dataset:

- If `album_release_date` exists, `year`, `month`, and `day` can be derived directly.
- Missing `featured_artists` and album-related metadata may be enriched using external music databases or APIs, queried by track title and primary artist.
- For tracks with available lyrics, additional text-based features (e.g., sentence count, token count, average tokens per clause) can be computed, and missing language

labels can be inferred.

1.6 Derived Information Opportunities

The datasets provide multiple opportunities for feature derivation:

- **Lyrics-based features:** token and sentence statistics, language detection, and content-based indicators.
- **Audio features:** tempo, energy, and spectral characteristics suitable for clustering or rule-based profiling.
- **Geographic features:** region-level aggregation and spatial analysis using H3 indexes.

These derived features enhance data quality and analytical value without requiring manual labeling and directly support later profiling and categorization tasks.

2 Assignment 2: Data Cleaning

2.1 Objective

The objective of Assignment 2 is to address the missing-data problems identified in Assignment 1 and to integrate additional data where feasible. Specifically, the goal is to (1) reduce missingness in fields required for subsequent analysis, (2) recover missing values using reasonable imputation or external enrichment, and (3) produce cleaned, analysis-ready datasets while documenting any remaining missing values and the reasons for keeping them.

2.2 Cleaning Strategy Overview

Based on Assignment 1 findings, the cleaning approach followed three principles:

1. Fix missing values only where meaningful and reliable (e.g., geographic fields using geocoding, derived metrics using lyrics).
2. Integrate external data when it improves dataset completeness (e.g., Spotify meta-data; spatial indexing via Uber H3).
3. Preserve *structural missingness* where missing values represent real-world absence (e.g., instrumental tracks have no lyrics).

2.3 Artists Dataset Cleaning (`artists.xml` → `Cleaned_Artists.csv`)

2.3.1 Geographic Enrichment

The primary cleaning work for the artists dataset focused on improving geographic completeness (`latitude`, `longitude`, `province`, `region`, `country`, `birth_place`). Forward geocoding was applied when `birth_place` was available but coordinates were missing, while reverse geocoding was used when coordinates existed but location text fields were incomplete. If geocoding results were unreliable or unavailable, the record was left unchanged to avoid introducing incorrect values.

Listing 2: Geocoding logic for recovering artist geographic information

```

1 # Forward geocoding
2 if birth_place exists and (latitude or longitude missing):
3     loc = geocode(birth_place)
4     if loc found:
5         fill latitude, longitude, country, region, province
6
7 # Reverse geocoding
8 elif latitude and longitude exist and admin fields missing:
9     loc = reverse_geocode(latitude, longitude)
10    if loc found:
11        fill missing country, region, province, birth_place

```

2.3.2 Rule-Based Nationality Completion

For missing `nationality`, a simple and transparent rule was applied: if `country` was available, `nationality` was set equal to `country`. This improves completeness without introducing ambiguity.

2.3.3 Spatial Index Integration (Uber H3)

Once coordinates were available (original or recovered), an H3 spatial index (`h3_index`) was computed at resolution 4. This enables hierarchical spatial grouping and region-level aggregation in later analysis.

2.3.4 Post-Cleaning Missingness Summary (Artists)

After cleaning, the artists dataset contains 104 records. Remaining missing values are limited to biographical fields (`birth_date` and `description`), while all geographic fields and the derived `h3_index` are complete. Table 3 summarizes the remaining missingness.

Table 3: Remaining missing values per field in the cleaned artists dataset

Field	Missing Count
birth_date	31
description	18

2.4 Tracks Dataset Cleaning (`tracks.json` → `Cleaned_Tracks.csv`)**2.4.1 Spotify Metadata Enrichment**

To reduce missingness in collaboration and album metadata, Spotify API enrichment was applied using *track title + primary artist* as the query. When a match was found, missing album-related fields were filled. Featured artists were derived by excluding the primary artist from the list of artists returned by Spotify.

Listing 3: Spotify enrichment logic for missing featured_artists and album metadata

```

1 # Query Spotify using: title + primary_artist
2 if featured_artists is missing:
3     track = spotify_search(title, primary_artist)
4     if track found:
5         fill album fields if missing (album_name/type/release_date,
6         duration, etc.)
7         all_artists = track.artists
8         featured = all_artists excluding primary_artist
9         if featured not empty:
10             featured_artists = join(featured)

```

2.4.2 Audio Feature Imputation (Artist Mean)

For missing audio features (e.g., BPM, loudness, spectral features), a within-artist mean imputation strategy was applied. For each artist, feature means were computed from available tracks and used to fill missing values, preserving artist-specific style characteristics.

2.4.3 Derived NLP Metrics from Lyrics

For tracks with available lyrics but missing NLP-derived metrics, these metrics were recomputed directly from the lyrics text. This ensures internal consistency by making derived variables deterministic functions of lyrics.

2.4.4 Language and Date Completion

Missing language labels were inferred using language detection on lyrics text. Missing year, month, and day values were derived from album_release_date. If only year-month information was available, day was set to 1 (documented assumption).

2.4.5 Fields Intentionally Left Unfilled

Some missing values were deliberately retained:

- `id_album` (78 missing): external album identifiers do not match the dataset's internal ID scheme.
- `lyrics` (3 missing): instrumental tracks; dependent NLP fields remain missing by design.

2.4.6 Post-Cleaning Missingness Summary (Tracks)

After cleaning, the tracks dataset contains 11,166 records. Remaining missingness is summarized in Table 4.

Table 4: Remaining missing values per field in the cleaned tracks dataset

Field	Missing Count
featured_artists	4240
n_sentences	3
n_tokens	3
char_per_tok	3
avg_token_per_clause	3
id_album	78
lyrics	3

3 Assignment 3: Song Profiling

3.1 Objective

The objective of this assignment is to build song categories using the information available in the cleaned dataset. Since melodic/audio attributes were available for most tracks, the categorization was based on audio features rather than lyrics, producing an interpretable set of song profiles that can be used for later analysis (e.g., grouping, comparisons, and aggregation).

3.2 Methodology

3.2.1 Data Used

The cleaned tracks dataset (`Cleaned_Tracks.csv`) was used as input. A new column, `song_category`, was created and saved into `Cleaned_Tracks(with_categories).csv`.

3.2.2 Feature Selection

Four melodic/audio features were selected because they represent two major perceptual dimensions of music:

- **BPM (tempo)**: indicates how fast the song feels.
- **RMS (energy)**: reflects overall intensity/energy of the audio signal.
- **Rolloff (spectral rolloff)**: approximates high-frequency content (“brightness”).
- **Loudness**: reflects perceived volume/production intensity.

Together, BPM and RMS capture *energy*, while rolloff and loudness capture *brightness/-calmness*, making them suitable for broad profiling.

3.2.3 Thresholding Strategy

Instead of using fixed thresholds (e.g., $\text{BPM} > 120$), thresholds were derived from the dataset distribution using the 40th and 60th percentiles. This makes the rules adaptive to the dataset and reduces sensitivity to outliers or genre-specific biases. It also creates a stable “mid-range” band and avoids overly skewed categories.

3.2.4 Category Rules

Each song is assigned to a category using two intermediate assessments:

(1) **Energy level (from BPM and RMS)**

- **High energy:** BPM > 60th percentile **OR** RMS > 60th percentile
- **Low energy:** BPM < 40th percentile **AND** RMS < 40th percentile
- **Mid energy:** otherwise

(2) **Brightness level (from rolloff and loudness)**

- **Bright:** rolloff > 60th percentile **OR** loudness > 60th percentile
- **Calm:** rolloff < 40th percentile **AND** loudness < 40th percentile
- **Neutral:** otherwise

Final song categories (4 groups)

- **Energetic/Party:** High energy + Bright
- **Energetic/Emotional:** High energy + (Neutral or Calm)
- **Soft/Chill:** Low energy + Calm
- **Mellow/Bright:** all remaining cases (mainly mid energy and/or neutral brightness)

If any required audio fields (BPM, RMS, rolloff, loudness) are missing for a track, the category is set to **Unknown** to avoid unreliable assignment.

Listing 4: Category Derivation Logic

```

1 if energy == "high" and bright == "bright":
2     return "Energetic/Party"
3 if energy == "high":
4     return "Energetic/Emotional"
5 if energy == "low" and bright == "calm":
6     return "Soft/Chill"
7 return "Mellow/Bright"

```

3.2.5 Results

The final category distribution is reported in Table 5. The dataset is dominated by energetic tracks, while **Soft/Chill** represents a smaller subset.

Table 5: Distribution of songs across the four profiling categories

Song category	Count
Energetic/Party	5440
Energetic/Emotional	1538
Soft/Chill	661
Mellow/Bright	3527

4 Assignment 4: Data Warehouse Schema

4.1 Objective

The primary objective of this assignment was to transition from the raw data preparation phase (Python) to a structured relational environment suitable for analytical processing. We aimed to design a logical **Star Schema** that optimizes query performance for aggregation and reporting, specifically enabling multidimensional analysis across Time, Geography, and Artist demographics.

4.2 Schema Design

We designed a **Star Schema** to facilitate efficient analytical queries. The schema consists of one Fact table and three Dimension tables.

4.2.1 Fact Table: Fact_Streams

This table captures the transactional events (streams).

- **Primary Key:** Composite key (`id_track`, `id_artist`, `id_time`).
- **Measures:**
 - `streams` (from `streams@1month`)
 - `popularity`

4.2.2 Dimension Tables

1. **Dim_Time:** Handles the temporal hierarchy.
Attributes: `id_time`, Year, Month, Day, Quarter, Season.
2. **Dim_Artist:** Stores artist demographics and location data.
Attributes: `id_artist`, Name, Gender, Country, Region, H3 Index, Lat/Lon.
3. **Dim_Track:** Stores song characteristics and profile data.
Attributes: `id_track`, Title, Song Category, Duration, Language, Explicit, Featured Artists, and Audio/NLP Metrics.

This schema allows us to slice streams by Time (Seasonality), Location (Geography/H3), and Content Type (Song Category).

5 Assignment 5: Data Preparation

5.1 Objective

The goal of this assignment was to transform the enriched, single-source datasets (tracks and artists) into a normalized structure compatible with our Data Warehouse schema. This involved splitting the data into four distinct CSV files (`Dim_Artist`, `Dim_Track`, `Dim_Time`, and `Fact_Streams`) while rigorously enforcing referential integrity to prevent insertion errors during the database loading phase.

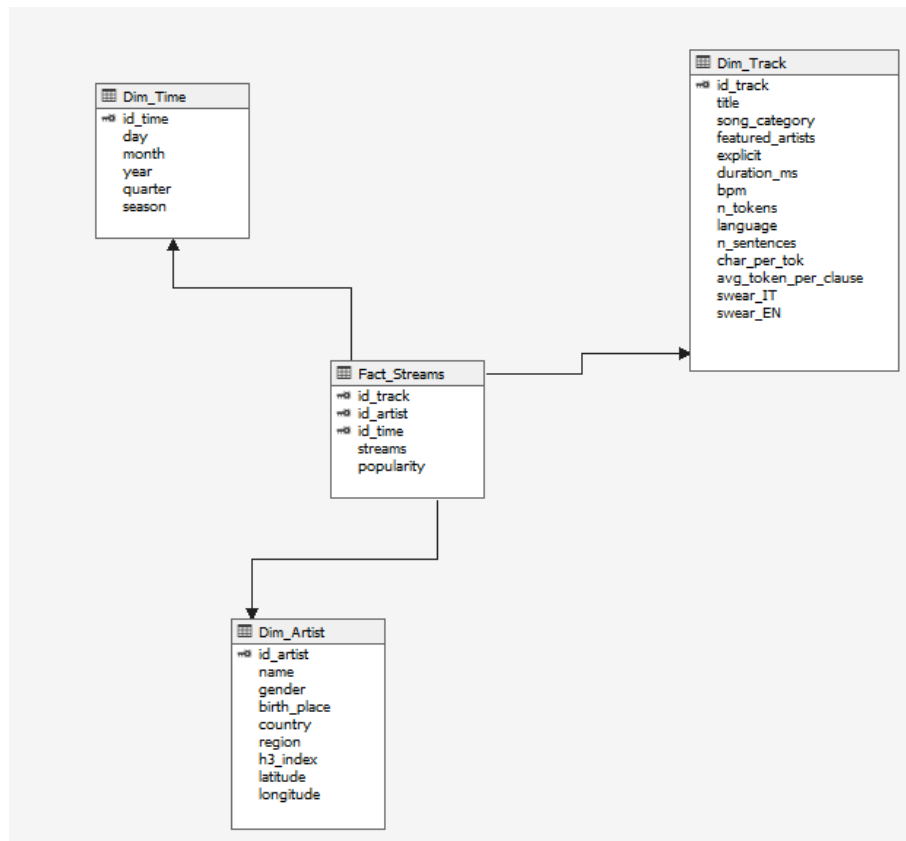


Figure 1: Star Schema Design: Fact Streams linked to Dimensions.

5.2 Splitting Strategy

We developed a Python script to split the fully processed dataset `Cleaned_Artists_.csv` and `Cleaned_Tracks_(with_categories)_.csv` into four separate CSV files corresponding to the DW Schema defined in Assignment 4.

5.3 Referential Integrity

To ensure data quality before database insertion, the script performs the following checks:

1. **Unique Identifiers:** Using Sets (`seen_track_ids`) to prevent duplicate dimension entries.
2. **Fact Integrity:** A row is added to `Fact_Streams` only if the `id_artist` exists in the valid artist list.
3. **Time Key Generation:** A generic `id_time` key (Format: YYYYMMDD) is generated to link facts to the Time dimension.

Listing 5: Integrity Check during Splitting

```

1 # Only add fact if artist exists and key is unique
2 fact_key = (row['id'], row['id_artist'], id_time)
3
4 if row['id_artist'] in valid_artist_ids and fact_key not in
   seen_fact_rows:
5     fact_rows.append({...})
6     seen_fact_keys.add(fact_key)

```

The output files are stored in the DW_Import/ directory.

5.4 Generated Output Samples

The following tables display a subset of the generated CSV files ready for bulk loading.

Table 6: Sample Data: Dim_Artist

ID Artist	Name	Gender	Birth Place	Country	Region	H3 Index	Lat	Long
ART82291002	99 Posse	M	Napoli	Italia	Campania	841e833fffffff	40.8359	14.2488
ART53496045	Achille Lauro	M	Verona	Italia	Veneto	841ea45fffffff	45.4425	10.9857
ART18853907	Alfa	M	Genova	Italia	Liguria	841f9b1fffffff	44.4073	8.9339
ART64265460	Anna Pepe	F	La Spezia	Italia	Liguria	841eb59fffffff	44.1034	9.8217
ART75741740	Articolo 31	M	Milano	Italia	Lombardia	841f99dfffffff	45.4642	9.1896
ART24123617	Babaman	M	Rho	Italia	Lombardia	841f99dfffffff	45.5289	9.0416

Table 7: Sample Data: Dim_Time

ID Time	Year	Month	Day	Quarter	Season
20210409	2021	4	9	2	Spring
20250516	2025	5	16	2	Spring
20200528	2020	5	28	2	Spring
20230209	2023	2	9	1	Winter
20220602	2022	6	2	2	Summer
20190607	2019	6	7	2	Summer
20201002	2020	10	2	4	Autumn
20190920	2019	9	20	3	Autumn

Table 8: Sample Data: Dim_Track

ID Track	Title	Category	Featured	Expl	Lang	Dur(ms)	BPM	Sent	Swear
TR934808	polska 2 :-/	Energetic/Party	Ernia; Guè	T	pl	207761	135.3	102	13
TR760029	POLKA	Energetic/Party	Thelonious B.	T	en	207761	129.4	56	9
TR916821	britney ;-)	Energetic/Party	MamboLosco...	T	en	193544	133.7	88	16
TR480968	CEO	Energetic/Emo	Taxi B	T	it	169000	162.2	37	8
TR585039	LONDRA	Mellow/Bright	Rkomi	T	en	194779	105.9	48	1

Table 9: Sample Data: Fact_Streams

ID Track	ID Artist	ID Time	Streams	Popularity
TR934808	ART04205421	20210409	186,522	46
TR760029	ART04205421	20210409	194,313	46
TR916821	ART04205421	20210409	63,750	39
TR480968	ART04205421	20250516	41,473	47
TR585039	ART04205421	20200528	30,553	41
TR550335	ART04205421	20200528	29,687	29
TR170793	ART04205421	20200528	24,103	23
TR627195	ART04205421	20200528	30,838	24
TR628871	ART04205421	20230209	20,522	50
TR700756	ART04205421	20200528	20,741	27
TR570242	ART04205421	20220602	14,401	40

6 Assignment 6: Data Uploading

6.1 Objective

The final step of the Data Preparation phase was to populate the Data Warehouse tables in the SQL Server environment. The objective was to develop a robust Python automation script that could connect to the remote server, parse the prepared CSV files, and efficiently bulk-load thousands of records into the **Fact** and **Dimension** tables while handling potential connection timeouts or data type conflicts.

6.2 Python-SQL Connection

We utilized the `pyodbc` library to connect to the SQL Server database (`YOUR_DATABASE`). The upload process was automated to handle the separate CSV files generated in the previous step.

6.3 Batch Loading

To optimize performance, data was uploaded in batches of 1,000 rows using `executemany`. This significantly reduced the I/O overhead compared to single-row inserts.

Listing 6: Batch Upload Function

```

1 def upload_table(cursor, table_name, file_name, sql_query, batch_size
  =1000):
2     # ... (reading file logic) ...
3     if len(batch_data) >= batch_size:
4         cursor.executemany(sql_query, batch_data)
5         batch_data = []

```

6.4 Execution Results

The script successfully populated the Data Warehouse. The performance metrics were as follows:

Table 10: Data Upload Performance

Table	Rows Uploaded	Time (s)
Dim_Time	1,377	7.64
Dim_Artist	104	0.65
Dim_Track	11,093	52.89
Fact_Streams	11,164	44.35

The database is now fully populated and ready for analytical processing.

Part II: ETL Implementation (SSIS)

SSIS Environment Configuration

Before commencing the ETL assignments (7–13), we configured the SQL Server Integration Services (SSIS) environment within Visual Studio. This setup was critical to ensure connectivity with the `remote server` and to standardize the execution flow.

Connection Managers

The project relies on two primary types of connection managers:

1. **OLE DB Connection:** Configured to connect to the SQL Server instance hosting our Data Warehouse.
 - **Server Name:** `YOUR_SQL_SERVER`
 - **Authentication:** SQL Server Authentication
 - **User:** `YOUR_SQL_USERNAME`
 - **Initial Catalog:** `YOUR_DATABASE`

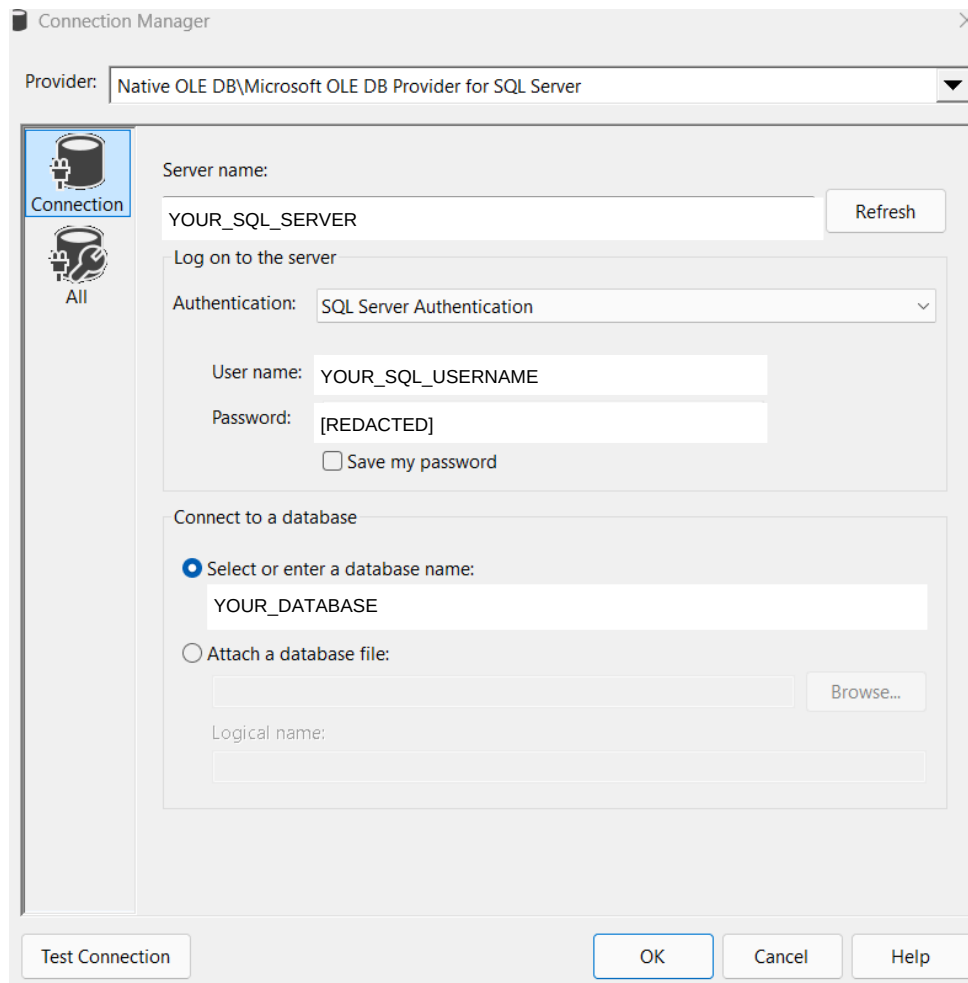


Figure 2: SSIS Connection Manager: Setup of OLE DB credentials and server connection.

2. **Flat File Connections:** Created dynamically for each assignment to export results as CSV files. We configured these with a comma delimiter (,) and text qualifiers (") to handle special characters in artist names.

Execution Context

The packages were developed and executed on the "Client Side" (Visual Studio environment). This aligns with the project constraint to minimize server-side computations. Data is extracted into the SSIS memory buffer, transformed using SSIS components (Aggregate, Sort, Lookup), and then loaded back or exported.

7 Assignment 7: Data Uploading (SSIS Sampling)

7.1 Objective

The primary objective of this assignment was to establish a lightweight development environment. Developing complex ETL transformations on the full dataset is computationally expensive and inefficient. To resolve this, we were required to:

1. Duplicate the Data Warehouse schema, appending the suffix `_SSIS` to all table names (e.g., `Fact_Streams_SSIS`).
2. Populate these tables with a random 30% sample of the original data using SQL Server Integration Services (SSIS).
3. **Crucial Requirement:** Maintain strict referential integrity. Every record in the sampled Fact table must point to an existing record in the sampled Dimension tables.

7.2 Sampling Strategy: Dimension-Driven Filtering

A naive approach to sampling (e.g., taking a random 30% of the Fact table) often leads to "Orphaned Facts"—streams that reference an Artist or Track that was excluded from the sample. To prevent this data integrity violation, we adopted a **Dimension-Driven Sampling Strategy**:

1. **Step 1:** We randomly sampled 30% of the *Dimensions* first.
2. **Step 2:** We loaded the *Fact Table* by filtering the full stream history against these reduced dimensions. Only facts belonging to the selected 30% of artists/tracks were loaded.

7.3 Package Implementation

The SSIS package `Assignment_07.dtsx` enforces this logic through a strictly ordered Control Flow. We utilized Precedence Constraints to ensure that Dimension tables are fully populated before the Fact table processing begins.

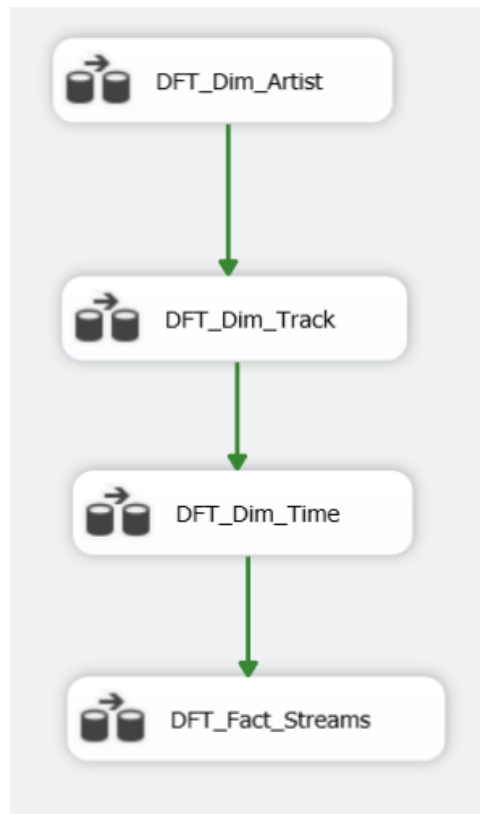


Figure 3: Control Flow: Enforcing load order (Dimensions → Facts) to ensure Referential Integrity

7.3.1 1. Dimension Sampling (Percentage Sampling)

For `Dim_Artist`, `Dim_Track`, and `Dim_Time`, we utilized the **Percentage Sampling** transformation. This component is highly efficient as it reduces the data volume early in the pipeline.

- **Input:** Full source table.
- **Logic:** Random seed with a 30% selection threshold.
- **Output:** The selected rows are inserted into the `_SSIS` destination tables. Unselected rows are discarded.

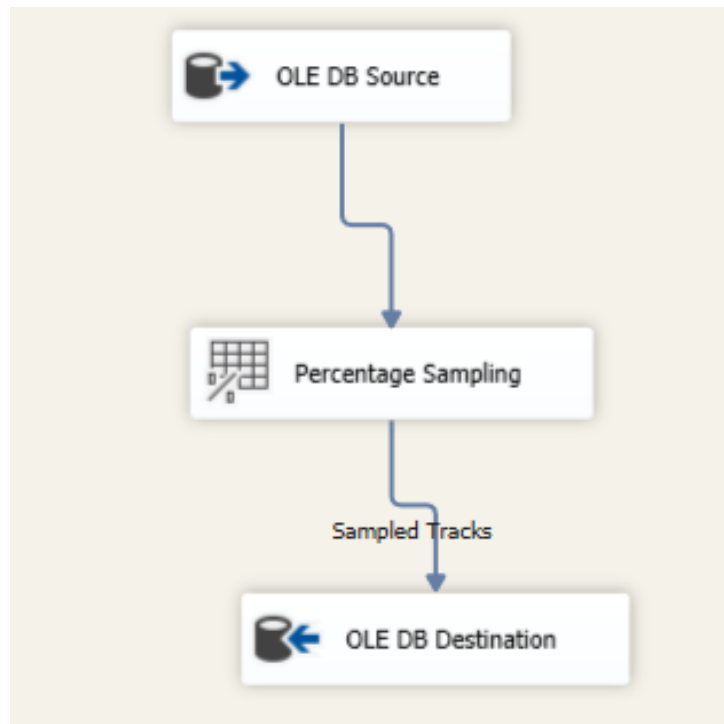


Figure 4: Data Flow Task for Dimension Sampling

7.3.2 2. Fact Table Filtering (Cascading Lookups)

The loading of `Fact_Streams_SIS` required a more sophisticated Data Flow. We could not use the "Percentage Sampling" component here because it ignores relationships. Instead, we used a cascade of Lookup Transformations acting as filters.

The Pipeline Logic:

1. **Source:** We read the full `Fact_Streams` table (100% of data).
2. **Lookup Artist:** The stream is checked against `Dim_Artist_SIS`.
 - *Match:* Proceed to next step.
 - *No Match:* The row is discarded (redirected to null/ignore).
3. **Lookup Track:** The surviving streams are checked against `Dim_Track_SIS`.
4. **Lookup Time:** The surviving streams are checked against `Dim_Time_SIS`.
5. **Destination:** Only rows that found a match in ALL three dimensions are inserted into `Fact_Streams_SIS`.

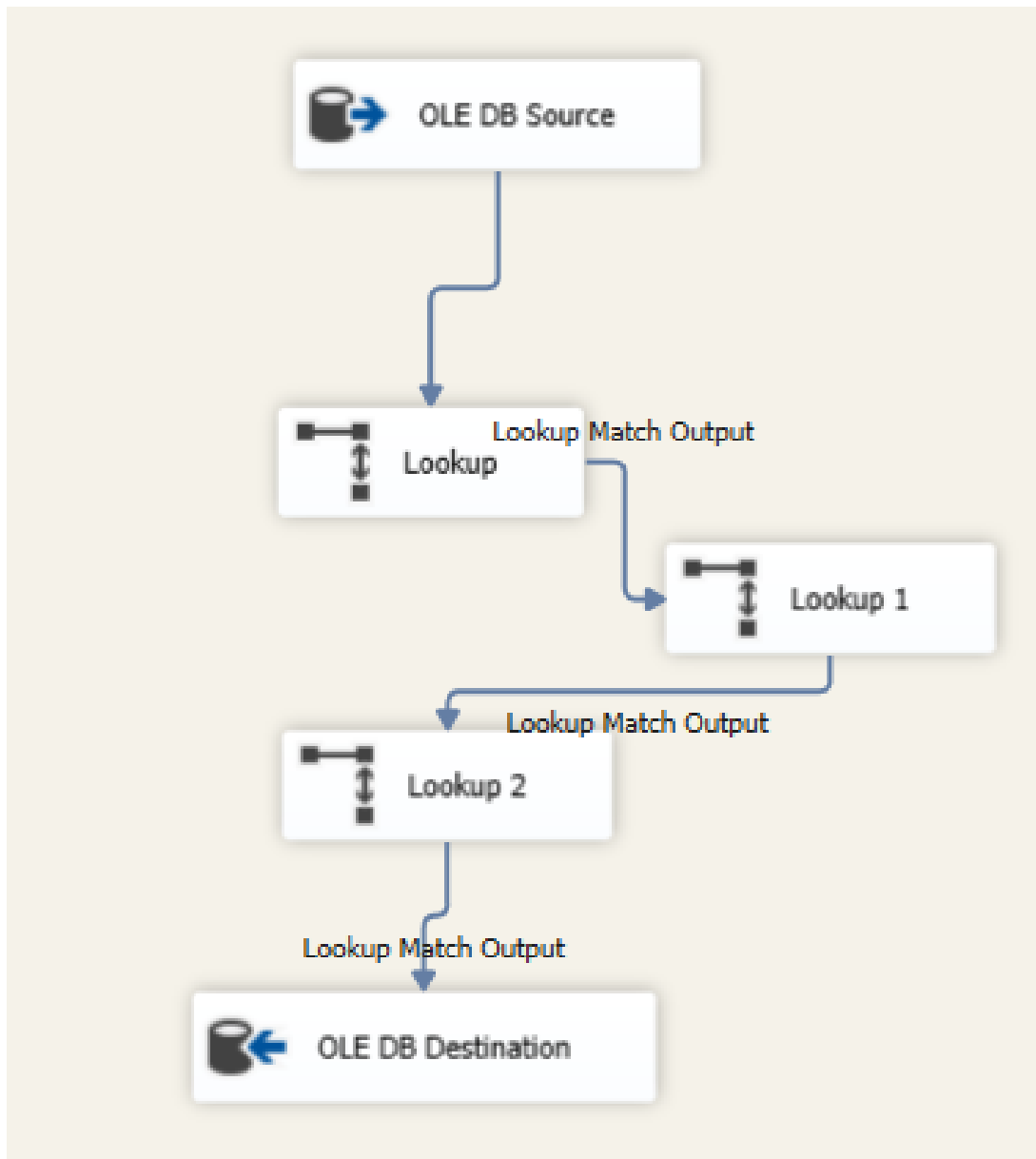


Figure 5: Fact Table Data Flow: Using Cascading Lookups to enforce integrity

7.4 Conclusion

This architecture successfully created a mathematically consistent subset of the data. By filtering the facts based on the sampled dimensions, we ensured that no foreign key constraints would be violated during the analysis phase. This significantly improved the performance of subsequent query development in SSIS.

8 Assignment 8: Temporal Artist Productivity Analysis

8.1 Objective

The goal of this assignment was to analyze artist productivity over time. Specifically, we were required to:

1. Aggregate the number of songs published by each artist for every year.
2. List the artists ordered by year (chronologically) and then by their total output (descending).
3. Perform this calculation using the SSIS Data Flow, minimizing SQL queries on the source database.

8.2 Implementation Strategy

We designed an SSIS package (`Assignment_08.dtsx`) that retrieves raw transactional data and enriches it with dimensional attributes through Lookups before aggregating.

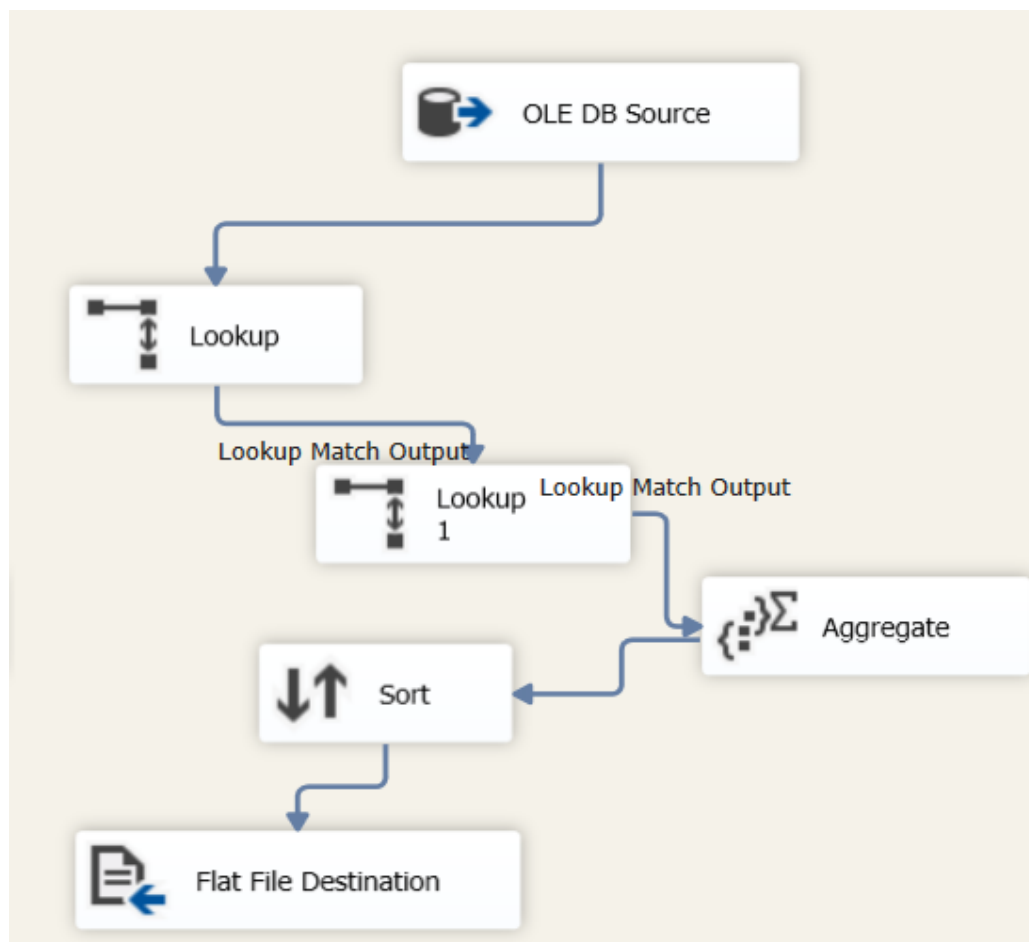


Figure 6: SSIS Data Flow for Artist Productivity Analysis: From lookup enrichment to aggregation.

8.2.1 1. Data Extraction & Enrichment

The process begins by reading the `Fact_Streams_SSIS` table. Since the Fact table contains only surrogate keys (`id_time`, `id_artist`), we needed to join it with dimensions to get human-readable attributes.

- **Lookup 1 (Time):** The stream's `id_time` is matched against `Dim_Time_SSIS` to retrieve the **Year** column.
- **Lookup 2 (Artist):** The stream's `id_artist` is matched against `Dim_Artist_SSIS` to retrieve the **Artist Name**.

Rows failing the lookup (if any) are redirected to error outputs to ensure data quality, though our sampling strategy in Assignment 7 ensures 100% matches.

8.2.2 2. Aggregation Logic

We utilized the **Aggregate Transformation** to compute the metrics.

- **Group By:** Year, Artist Name, `id_artist`.
- **Operation:** Count Distinct on `id_track`.

Justification for Count Distinct: The stream data represents "events." An artist might have thousands of streams for the same song in the same year. We are interested in the number of *unique songs* published/active, not the number of streams. Therefore, counting distinct Track IDs is the correct metric for "songs published."

8.2.3 3. Sorting & Output

To meet the reporting requirement ("ordered by total number of songs"), we used the **Sort Transformation**.

1. **Primary Sort Key:** Year (Ascending).
2. **Secondary Sort Key:** TotalSongs (Descending).

This ordering effectively produces a "Top Artists per Year" leaderboard. The final dataset is written to a CSV file named `Query_8_Results.csv`.

8.3 Sample Results

Table 11 presents a subset of the output data, demonstrating the correlation between the release year and artist productivity (defined as unique tracks released).

Table 11: Sample Output: Artist productivity grouped by year.

Year	Artist Name	Total Songs
1993	Articolo 31	2
1996	Articolo 31	2
1997	Articolo 31	7
1998	Articolo 31	6
1999	Il Tre	1
1999	Nerone	1
2002	Articolo 31	4
2003	Articolo 31	7
2005	Gemitaiz	1
2006	Babaman	5
2007	Cor Veleno	4

9 Assignment 9: Seasonal Regional Analysis (Summer vs. Winter)

9.1 Objective

The objective of this assignment was to perform a comparative analysis of seasonal music consumption across different geographical regions. Specifically, we were required to compute a "Summer-Winter Score" for each region, defined as the ratio of activity (songs released and total streams) in Summer versus Winter.

$$\text{Score} = \frac{\text{Metric}_{\text{Summer}}}{\text{Metric}_{\text{Winter}}}$$

A score > 1 indicates a region prefers Summer releases, while a score < 1 indicates a Winter preference.

9.2 Implementation Strategy

This query required separating the data into two distinct cohorts (Summer and Winter) and then rejoining them to calculate the ratio. We implemented this using a Split-Aggregate-Merge pattern in SSIS.

9.2.1 1. Data Enrichment & Cleaning

The pipeline begins by reading the `Fact.Streams.SSIS` table.

- **Lookups:** We joined with `Dim.Time.SSIS` to retrieve the `Season` attribute and `Dim.Artist.SSIS` to retrieve the `Region`.
- **Conditional Split (Filter Region):** To ensure data quality, we applied a filter `!ISNULL(Region) && LEN(Region) > 0`. This removes streams with missing geographical data, ensuring our regional scores are accurate.

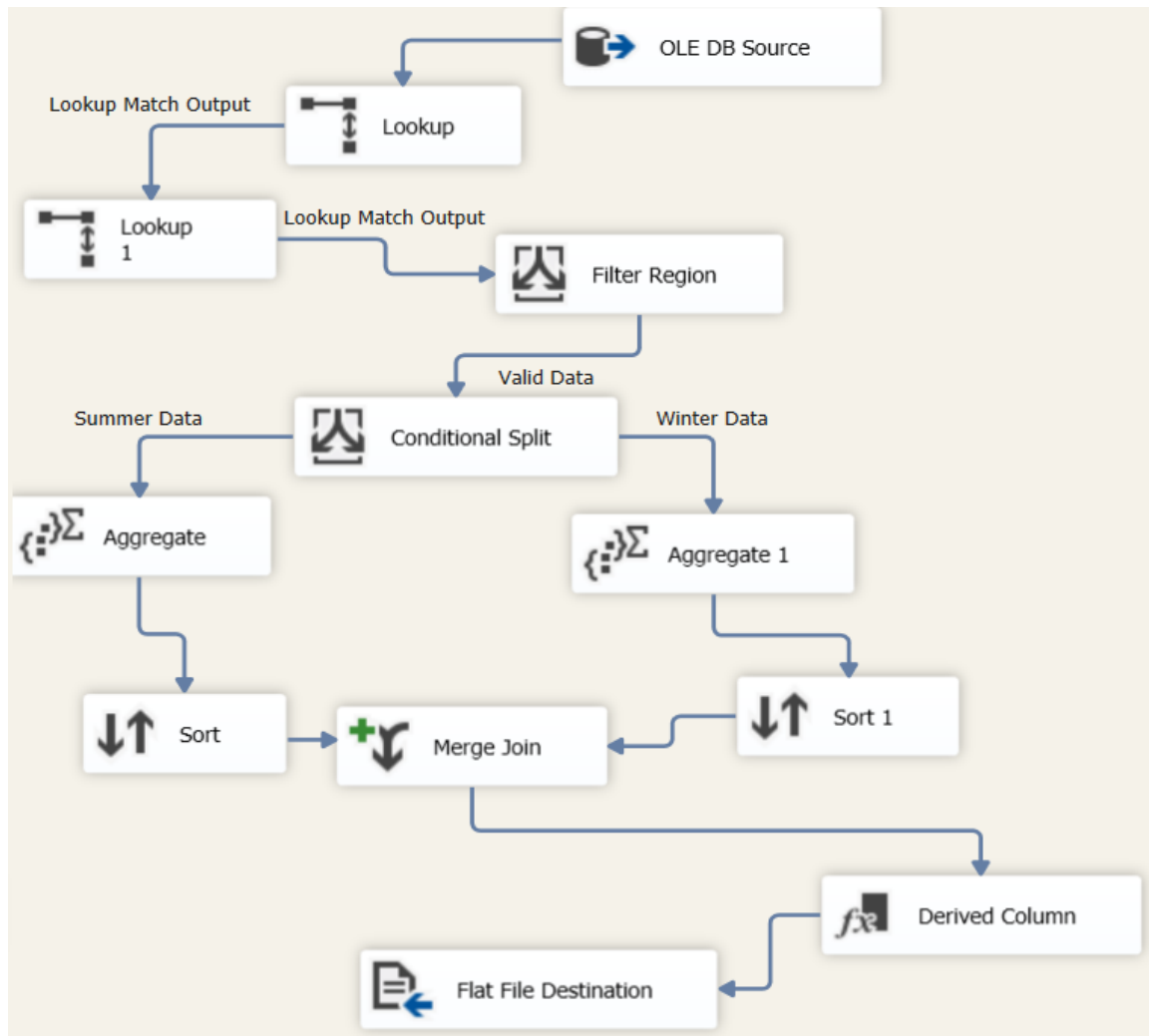


Figure 7: SSIS Data Flow: Split-Aggregate-Merge Pattern for Seasonal Analysis.

9.2.2 2. Seasonal Splitting

We used a **Conditional Split Transformation** to route rows into parallel processing paths based on the season:

- **Path 1 (Summer):** Condition Season == "Summer"
- **Path 2 (Winter):** Condition Season == "Winter"
- **Default Output:** Ignored (Spring/Autumn data is dropped from this specific analysis).

9.2.3 3. Parallel Aggregation

Each path flows into its own **Aggregate Transformation**.

- **Group By:** Region
- **Metrics:**

1. `Count Distinct(id.artist)` → To measure active artists.
2. `Sum(streams)` → To measure total consumption.

9.2.4 4. Rejoining (Merge Join)

SSIS requires sorted inputs for a Merge Join. We added **Sort Transformations** on the **Region** column for both the Summer and Winter paths. We then used a **Merge Join (Inner Join)** to combine the datasets. *Note:* Regions that only had activity in one season (but not the other) would be excluded by an Inner Join. This is acceptable for a "Ratio" calculation, as division by zero or division of zero would be mathematically undefined or trivial.

9.2.5 5. Score Calculation (Derived Column)

Finally, we used a **Derived Column Transformation** to compute the ratios. We cast the values to `DT_NUMERIC` (Numeric with precision) to ensure floating-point division was performed instead of integer division.

Expression Used:

```
(DT_NUMERIC, 12, 4) ( (DT_R8)SummerStreams /
(DT_R8)WinterStreams )
```

9.3 Sample Results

Table 12 displays the calculated scores. For example, **Liguria** shows a very high preference for Summer streaming (Score 21.87), whereas **Lazio** shows a significant preference for Winter (Score 0.18).

Table 12: Seasonal Regional Analysis: Comparison of Summer vs. Winter activity. A score > 1 favors Summer.

Region	Summer Metrics		Winter Metrics		Seasonal Scores	
	Songs	Streams	Songs	Streams	Song Score	Stream Score
Campania	5	328,052	3	192,408	1.6666	1.7049
Lazio	3	72,058	23	388,977	0.1304	0.1852
Liguria	20	1,036,291	3	47,376	6.6666	21.8737
Lombardia	42	732,818	43	526,628	0.9767	1.3915
Piemonte	9	49,420	5	127,801	1.8000	0.3866
Veneto	3	65,000	1	5,211	3.0000	12.4736

10 Assignment 10: Local vs. Global Music Consumption

10.1 Objective

This assignment aimed to quantify the "hometown advantage" for artists. Specifically, for each song category and for each region (corresponding to the artist's birthplace), we

needed to compute the ratio:

$$\text{Ratio} = \frac{\text{Local Streams (Birthplace)}}{\text{Global Streams (Rest of World)}}$$

A high ratio suggests a genre or artist scene is locally focused, while a low ratio suggests broad international appeal.

10.2 Implementation Strategy

The complexity here lies in comparing a specific slice of data (Region) against the whole dataset (Global) simultaneously. We implemented a Multicast-Aggregate-Join pattern to calculate the numerator and denominator in parallel.

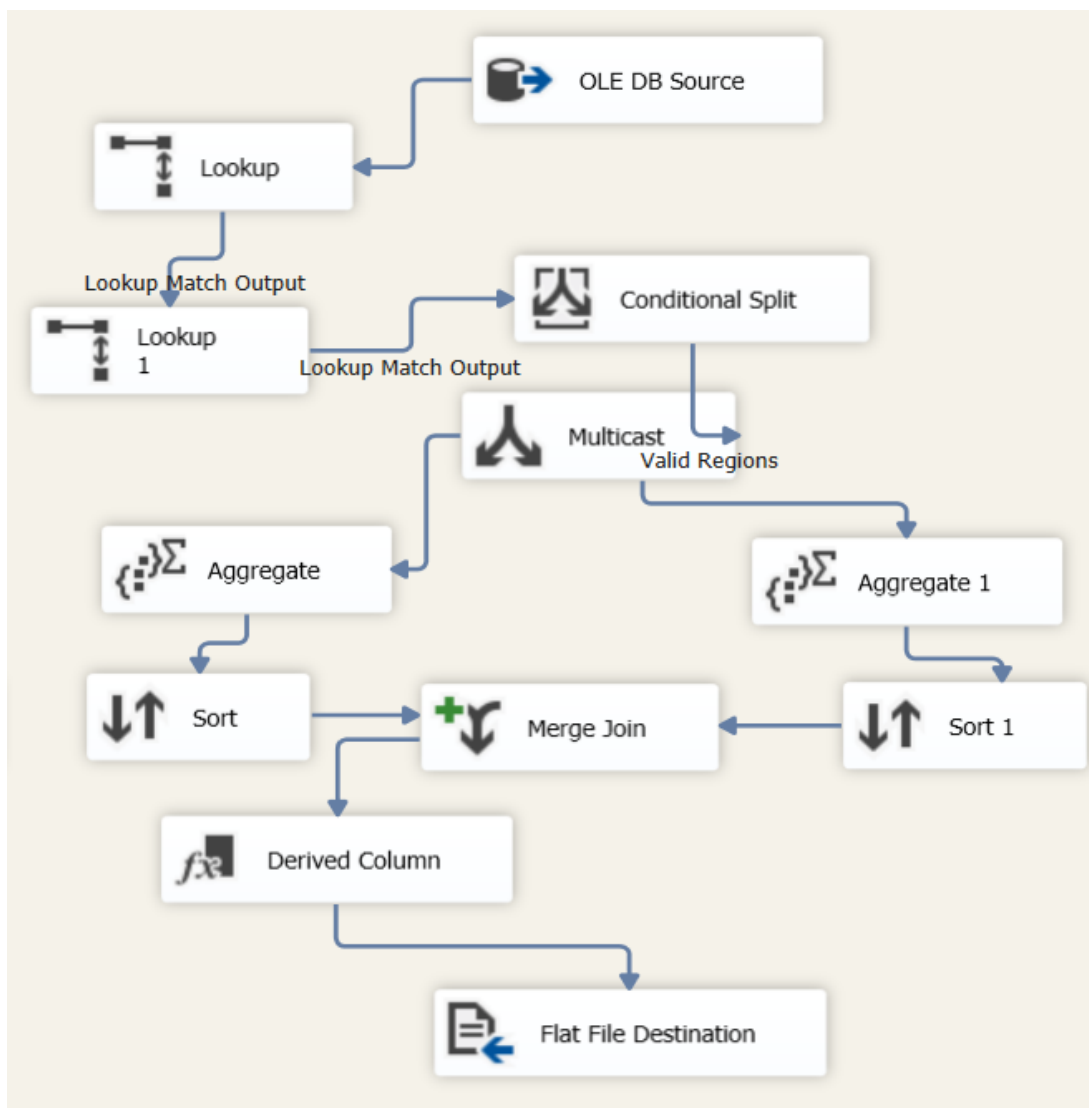


Figure 8: SSIS Data Flow: Calculating Local vs. Global totals via Multicast.

10.2.1 1. Data Preparation

The pipeline starts by reading `Fact.Streams.SSIS`.

- **Lookups:** We joined with `Dim_Artist_SSIS` to get the `Region` and `Dim_Track_SSIS` to get the `Song_Category`.
- **Filtering:** We used a **Conditional Split** to remove records where `Region` was `NULL`. Analysis of "Local" impact is impossible without a known location.

10.2.2 2. Parallel Aggregation (Multicast)

We utilized a **Multicast Transformation** to duplicate the data stream into two branches:

Branch A: Local Totals (Numerator)

- **Group By:** `Region`, `Song_Category`
- **Metric:** `Sum(Streams)` as `LocalStreams`.
- *Result:* The volume of a specific genre in a specific region (e.g., "Pop in Lombardia").

Branch B: Global Category Totals (Denominator Base)

- **Group By:** `Song_Category`
- **Metric:** `Sum(Streams)` as `TotalCategoryStreams`.
- *Result:* The total volume of that genre worldwide (e.g., "Pop Total").

10.2.3 3. Rejoining and Calculation

We sorted both branches by `Song_Category` and used a **Merge Join (Inner Join)** to bring them together. This effectively appends the "Global Total" column to every "Local" row.

Finally, a **Derived Column** calculated the ratio. The formula required careful logic to adhere to the definition "all *other* regions":

$$\text{Denominator} = \text{TotalCategoryStreams} - \text{LocalStreams}$$

$$\text{Ratio} = \frac{\text{LocalStreams}}{\text{Denominator}}$$

Expression Used:

```
(DT_NUMERIC, 12, 4) ( (DT_R8)LocalStreams /  
( (DT_R8)TotalCategoryStreams - (DT_R8)LocalStreams ) )
```

10.3 Sample Results

Table 13 illustrates the results for the "Energetic/Emotional" category. For instance, **Liguria** shows a very high local concentration (Ratio 0.4404), indicating nearly half as many streams happened locally as happened in the rest of the world combined.

Table 13: Local vs. Global Consumption: Ratio calculation for ‘Energetic/Emotional’ tracks.

Region	Song Category	Local Streams	Total Category Streams	Ratio
Campania	Energetic/Emotional	289,027	1,542,229	0.2306
Lazio	Energetic/Emotional	399,499	1,542,229	0.3496
Liguria	Energetic/Emotional	471,543	1,542,229	0.4404
Lombardia	Energetic/Emotional	308,891	1,542,229	0.2504
Piemonte	Energetic/Emotional	35,095	1,542,229	0.0232
Sardegna	Energetic/Emotional	5,629	1,542,229	0.0036
Veneto	Energetic/Emotional	32,545	1,542,229	0.0215

11 Assignment 11: Trending Statistics & Featured Artist Analysis

11.1 Objective

This assignment required advanced statistical analysis to identify “Trending” and “Flopping” songs relative to an artist’s typical performance.

1. **Trending:** Streams $> (\mu + \sigma)$
2. **Flopping:** Streams $< (\mu - \sigma)$
3. **Metrics:** Compute the *Trending Percentage* and a normalized *Trending Factor* (-1 to $+1$).
4. **Featured Artists:** Perform the same analysis for featured artists, attributing the main artist’s trend status to them.

11.2 Implementation Strategy

We implemented a **Self-Join Architecture** to compare individual rows against aggregate statistics, followed by a **Script Component** to handle the one-to-many relationship of featured artists.

11.2.1 1. Statistical Calculation (The Self-Join)

Standard Deviation is not a native aggregate function in SSIS. We calculated it manually using the formula $\sigma = \sqrt{E[x^2] - (E[x])^2}$.

1. **Multicast 1:** Split the raw stream data into two paths.
2. **Path A (Aggregator):**
 - Derived Column: Calculated $Streams^2$.
 - Aggregate: Grouped by `id_artist` to get $\sum Streams$, $\sum Streams^2$, and Count (N).
 - Derived Column: Computed μ and σ .



Figure 9: SSIS Data Flow: Statistical Benchmarking and Featured Artist Parsing.

3. **Path B (Detail):** Sorted raw data to prepare for join.

4. **Merge Join:** Joined Path A and Path B on `id_artist`. This effectively appended the artist's average and standard deviation to every single track row.

11.2.2 2. Flagging Logic

With the statistics attached, we used a Derived Column to flag each track.

Listing 7: Trending Logic Expression

```

1 // Trending Flag (1 or 0)
2 (Streams > (Mean + StdDev)) ? 1 : 0
3
4 // Flopping Flag (1 or 0)
5 (Streams < (Mean - StdDev)) ? 1 : 0

```

Edge Case Handling: For artists with only $N = 1$ song, the Standard Deviation is 0. The condition becomes $Streams > Streams$, which is always False. This correctly results in 0 Trending and 0 Flopping songs, assigning a neutral Factor of 0. This avoids the mathematical error of dividing by zero standard deviation.

11.2.3 3. Featured Artist Parsing (Script Component)

To analyze featured artists, we needed to normalize the `featured_artists` column (e.g., "Drake, Future" $\rightarrow$ 2 rows). We used a Script Component (C#) acting as an asynchronous transformation.

Listing 8: C# Script for Splitting Featured Artists

```

1 public override void Input0_ProcessInputRow(Input0Buffer Row)
2 {
3     if (!Row.featuredartists_IsNull)
4     {
5         string[] artists = Row.featuredartists.Split(new char[] { ',' },
6             StringSplitOptions.RemoveEmptyEntries);
7         foreach (string artist in artists)
8         {
9             Output0Buffer.AddRow();
10            Output0Buffer.Featured_Artist = artist.Trim();
11            // Attribute the Main Artist's success to the Featured
12            Artist
13            Output0Buffer.Trend_Flag = Row.IsTrending;
14            Output0Buffer.Flop_Flag = Row.IsFlopping;
15        }
16    }
17 }

```

11.2.4 4. Final Aggregation

We performed two final aggregations in parallel:

- Main Artists: Aggregated flags by `id_main_artist`.
- Featured Artists: Aggregated flags by `Featured_Artist`.

The Trending Factor was calculated as:

$$\text{Factor} = \frac{\text{Count}_{\text{Trend}} - \text{Count}_{\text{Flop}}}{\text{Count}_{\text{Trend}} + \text{Count}_{\text{Flop}}}$$

(With logic to handle division by zero if $\text{Total} = 0$).

11.3 Sample Results

The following tables illustrate the calculated statistics. Table 14 shows the impact of featured artists, while Table 15 details the performance of main artists.

Table 14: Featured Artist Analysis: Attribution of trending status.

Featured Artist	Trends	Flops	Total	Main Artist ID	Trend %	Factor
Mostro	0	0	1	ART08177154	0.0000	0.0000
Emis Killa	0	1	1	ART85780419	0.0000	-1.0000
Tony Effe	0	0	1	ART05528539	0.0000	0.0000
Yung Snapp	0	0	1	ART85046033	0.0000	0.0000
Ensi	0	0	1	ART19729064	0.0000	0.0000
Lazza	0	0	1	ART86549066	0.0000	0.0000
Night Skinny	1	0	1	ART89596800	1.0000	1.0000
Juli	0	0	1	ART52465778	0.0000	0.0000
CaneSecco	0	0	2	ART85046033	0.0000	0.0000

Table 15: Main Artist Statistics: Calculated from self-join benchmarking.

Artist ID	Trends	Flops	Total	Trend %	Factor
ART61734477	1	0	7	0.1428	1.0000
ART75741740	1	3	28	0.0357	-0.5000
ART59609037	1	1	7	0.1428	0.0000
ART20729624	1	1	3	0.3333	0.0000
ART57587384	3	0	15	0.2000	1.0000
ART67409252	0	0	2	0.0000	0.0000
ART89596800	3	0	23	0.1304	1.0000

12 Assignment 12: Custom Business Query - Gender Performance Analysis

12.1 Objective

Business Question: "Is there a performance disparity in average streaming numbers across different artist genders?"

This query was chosen to provide insights into demographic trends within the music catalog. Understanding which demographic segments drive the highest engagement (streams per song) helps record labels target their A&R (Artists and Repertoire) efforts more effectively.

12.2 Implementation Strategy

We designed a pipeline to aggregate total streams and song counts by gender, handling data quality issues (missing gender values) proactively.

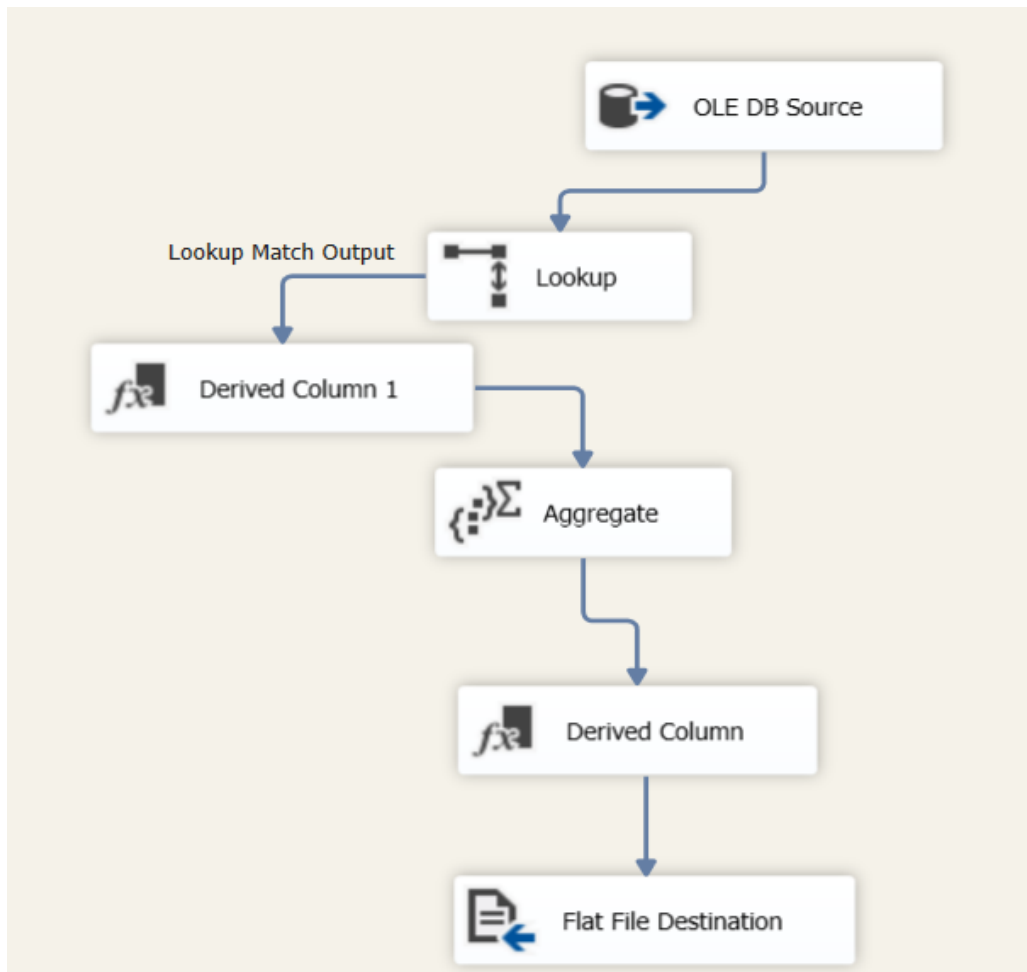


Figure 10: SSIS Data Flow for Gender Performance Analysis.

12.2.1 1. Data Extraction & Enrichment

The pipeline reads from Fact_Streams_SIS and performs a Lookup against Dim_Artist_SIS to retrieve the `gender` column.

12.2.2 2. Data Cleaning (Derived Column)

A significant portion of the artist data had missing or NULL gender values. Simply dropping these rows would skew the "Total Streams" calculation for the platform. Instead, we implemented a **Derived Column Transformation** to categorize them.

Listing 9: Expression for Handling Missing Gender

```
1 ISNULL(gender) || LEN(TRIM(gender)) == 0 ? "Others" : gender
```

This logic checks if the value is NULL *or* an empty string (whitespace). If either is true, it assigns the label "Others"; otherwise, it keeps the original value (Male/Female/Mixed/etc.).

12.2.3 3. Aggregation

We used the **Aggregate Transformation** to group data by the cleaned `Gender` column. We calculated two metrics:

- **Total Streams:** `Sum(streams)`
- **Song Count:** `Count(*)` (mapped as `totalSongs`)

12.2.4 4. Performance Metric Calculation

A final Derived Column calculated the key performance indicator (KPI): **Average Streams per Song**.

$$\text{Avg_Streams} = \frac{\text{Total Streams}}{\text{Total Songs}}$$

We cast the result to `DT_NUMERIC(12,2)` to ensure a readable, two-decimal format for the report.

12.3 Sample Results

Table 16 illustrates the performance disparity found in the dataset. While male artists account for the bulk of the volume (both songs and total streams), they also maintain a significantly higher average engagement per song compared to female artists in this specific dataset.

Table 16: Gender Performance Analysis: Average streams per song by gender category.

Gender	Total Streams	Total Songs	Avg Streams/Song
F	152,408	23	6,626.43
M	5,478,385	275	19,921.40

13 Assignment 13: Consultancy for Record Label - Content Strategy

13.1 Objective

Scenario: The streaming company is providing consultancy to a Record Label. The label wants to know if producing "Explicit" content restricts their audience or drives higher engagement per track compared to "Clean" (non-explicit) content.

Business Question: "Does the 'Explicit' tag correlate with higher average streaming numbers per song?"

13.2 Implementation Strategy

We designed a pipeline to categorize the entire catalog into "Explicit" or "Clean" cohorts and compare their performance efficiency.

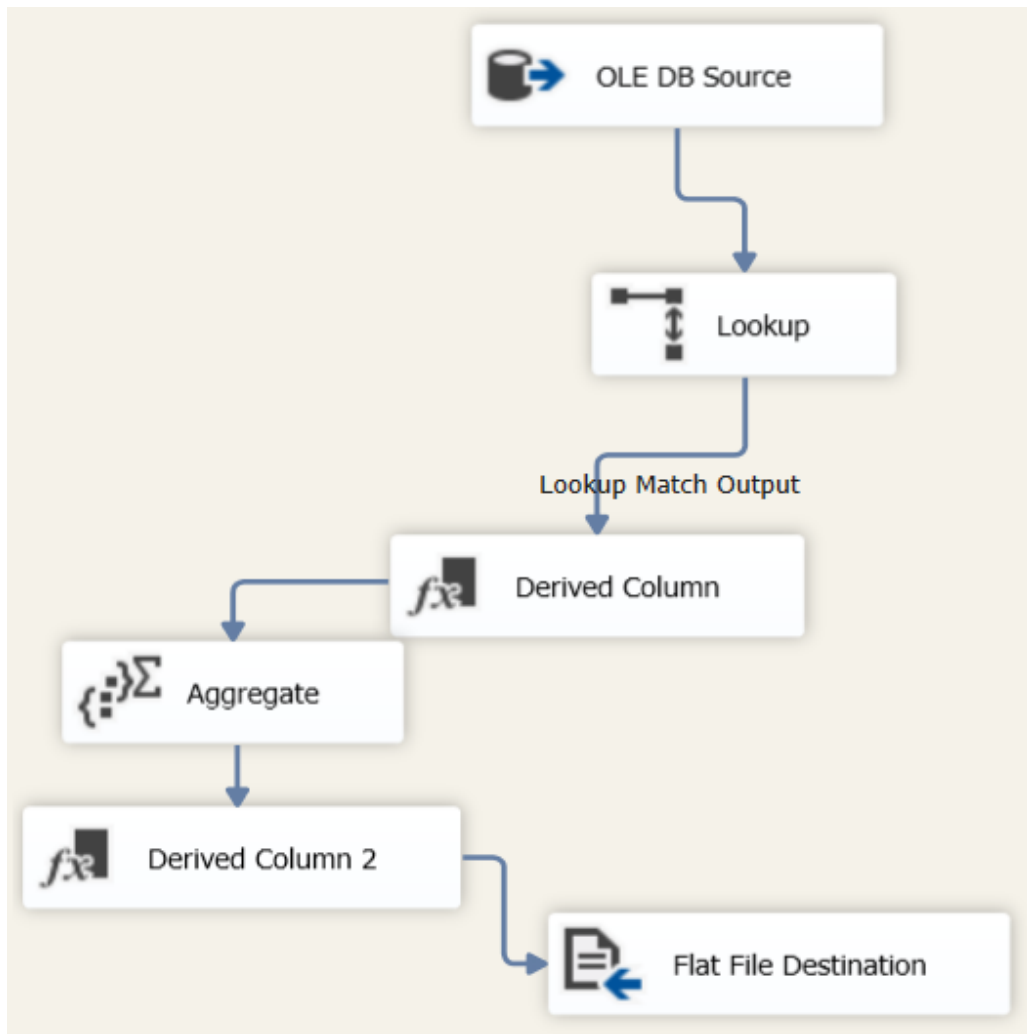


Figure 11: SSIS Data Flow for Explicit Content Analysis.

13.2.1 1. Data Normalization (Handling Mixed Types)

The pipeline begins by joining `Fact_Streams_SSIS` with `Dim_Track_SSIS`. Upon inspection, the `explicit` column contained heterogeneous data types (boolean bits, string "True"/"False", and integer "1"/"0") as well as NULLs.

To perform an accurate aggregation, we standardized this using a **Derived Column Transformation** with a nested conditional expression.

Listing 10: Logic to Standardize Explicit Status

```

1 // IF Null -> Clean
2 // IF "True" or "1" -> Explicit
3 // ELSE -> Clean
4
5 ISNULL(explicit) ? "Clean" :
6 (explicit == "True" || explicit == "1") ? "Explicit" : "Clean"

```

This ensures that every single track is definitely categorized, preventing data loss during the aggregation phase.

13.2.2 2. Comparative Aggregation

We used the **Aggregate Transformation** to group by the new `Content_Type` column.

- **Total Volume:** `Sum(Streams)`
- **Catalog Size:** `Count(*)` (Total Songs)

13.2.3 3. Efficiency Calculation

A record label cannot simply look at "Total Streams" because there are likely many more "Clean" songs than "Explicit" ones in the database. To make a fair comparison, we calculated the Average Streams per Song using a Derived Column.

$$\text{Avg_Streams} = \frac{\text{Total Streams}}{\text{Total Songs}}$$

13.3 Business Insight & Results

The results, exported to `Assignment13_ExplicitAnalysis.csv`, provide a clear recommendation. As shown in Table 17, "Clean" content actually outperforms "Explicit" content significantly in this dataset (approx. 24k vs 15k streams per song). This suggests that for this specific market, broader accessibility (radio-friendly content) may drive higher engagement than explicit content.

Table 17: Content Strategy Analysis: Performance of Explicit vs. Clean tracks.

Content Type	Total Songs	Total Streams	Avg Streams/Song
Explicit	163	2,393,125	14,681.75
Clean	135	3,237,668	23,982.73

Part III: Multidimensional Analysis (SSAS)

SSAS Configuration & Environment Setup

Before constructing the OLAP Cube, we configured the SQL Server Analysis Services (SSAS) environment within Visual Studio (SQL Server Data Tools). This setup was necessary to establish a connection between our local development environment and the remote server .

Data Source Configuration

We created a new Analysis Services Multidimensional Project and defined a **Data Source** to connect to the Data Warehouse.

- **Provider:** Native OLE DB\SQL Server Native Client 11.0
- **Server Name:** YOUR_SQL_SERVER
- **Authentication:** SQL Server Authentication (User: YOUR_SQL_USERNAME). We enabled "Save my password" to allow the SSAS service to authenticate during processing.
- **Impersonation Info:** Configured to *Inherit*, ensuring the service account could read from the relational database.

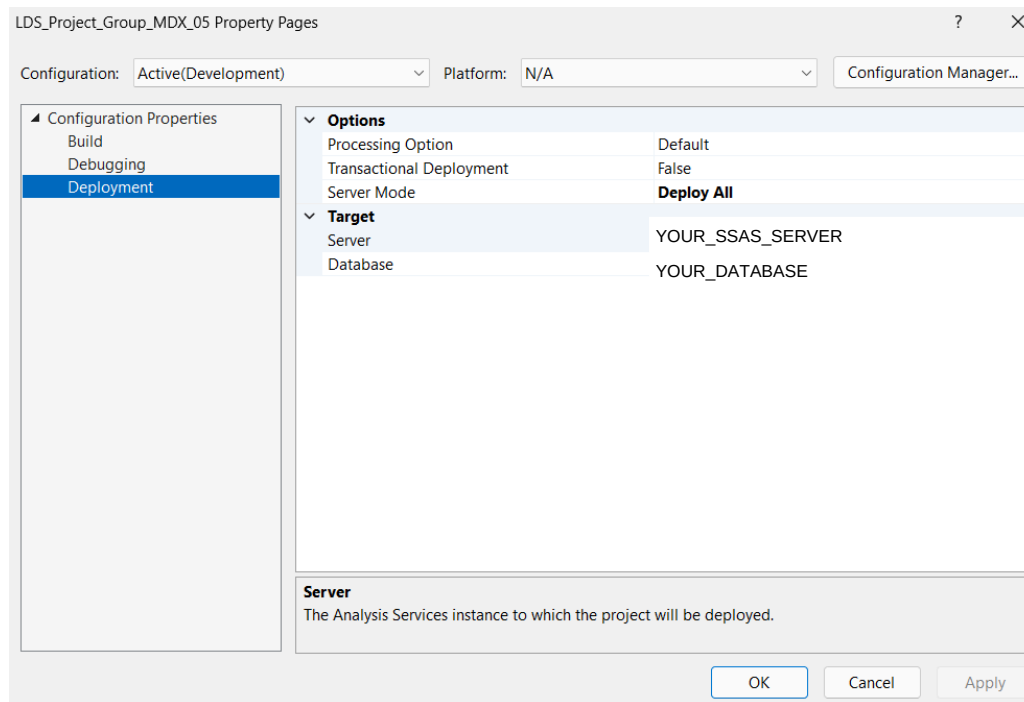


Figure 12: SSAS Data Source Configuration: Server authentication and impersonation details.

Deployment Properties

To deploy the cube to the remote server, we modified the project properties:

- **Target Server:** YOUR_SQL_SERVER
- **Target Database:** YOUR_DATABASE
- **Deployment Mode:** *Deploy All* (Ensures dimensions, cubes, and roles are updated).

14 Assignment 14: Data Cube Construction (SSAS)

14.1 Objective

The transition from relational database queries (SQL) and ETL processes (SSIS) to multi-dimensional analysis required the creation of an OLAP Cube using Microsoft SQL Server Analysis Services (SSAS). The goal was to build a structure that supports fast aggregation, drill-down capabilities, and complex calculations (MDX) across our defined dimensions.

14.2 Cube Architecture

We created a new Analysis Services project and defined a **Data Source View (DSV)** based on our Data Warehouse schema (YOUR_DATABASE).

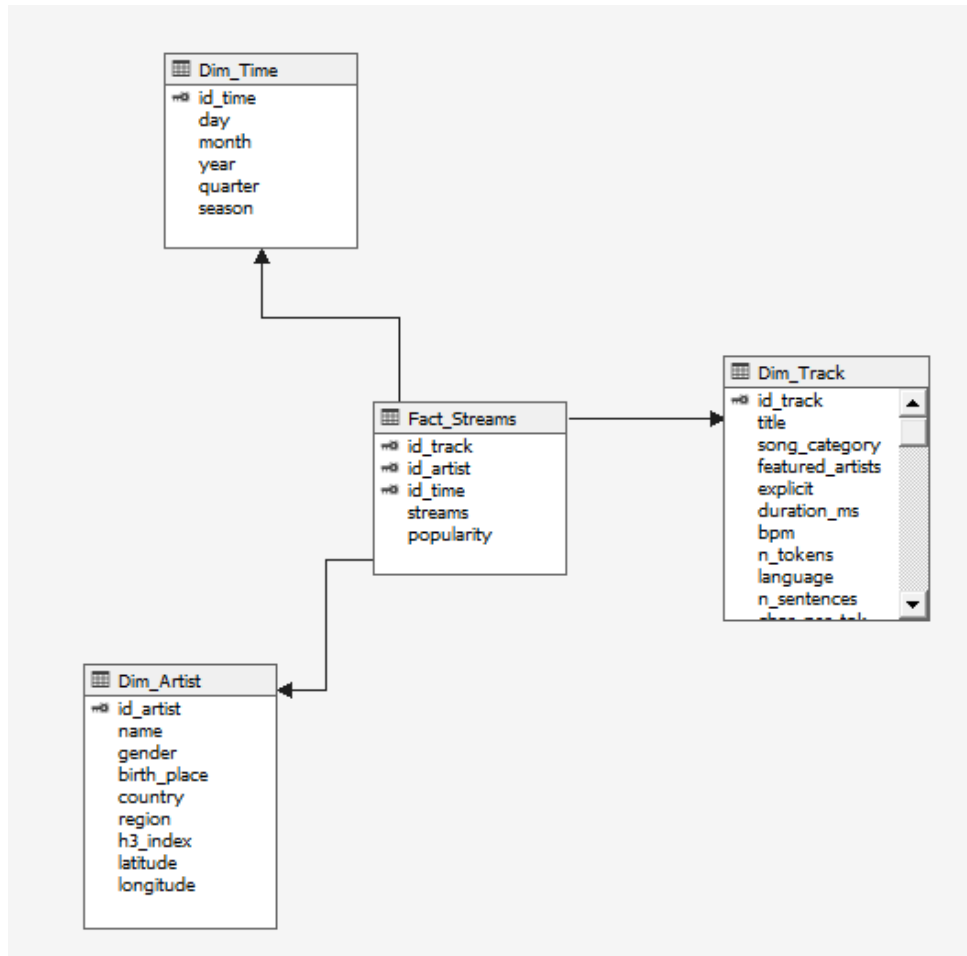


Figure 13: Star Schema representation in SSAS Data Source View

14.2.1 Measure Group

A single measure group was defined based on `Fact.Streams`:

- **Streams** (Sum): Primary metric for consumption analysis.
- **Popularity** (Sum / Average): Used for trending and benchmarking.
- **Fact Count** (Count): System-generated measure representing transaction volume.

14.2.2 Dimension Design and Hierarchies

To support efficient slice-and-dice operations, dimensions were processed and enriched with user-defined hierarchies.

- **Dim Time**: A standard calendar hierarchy was created to support temporal drill-down:

Year → Quarter → Month → Day

Attribute relationships were defined as `Day → Month → Quarter → Year` to enforce rigid aggregation paths. Although `Season` is available as an attribute, it was not included in the main hierarchy.

- **Dim Artist:** A geographical hierarchy was created to support regional analysis:

Country → Region → Id Artist

Attribute relationships were configured such that **Id Artist** relates to **Region**, and **Region** relates to **Country**. **Birth Place** was modeled as a descriptive property of the region.

- **Dim Track:** Tracks were organized by content category to support genre-level analysis:

Song Category → Track

Text-mining attributes were grouped into a display folder named *Lyrical Metrics*, containing fields such as **Swear IT**, **Swear EN**, **n Sentences**, and **Explicit**.

14.3 Processing & Deployment

The cube was deployed to the Analysis Services instance on the `remote server`.

Technical Challenge: During deployment, we encountered authentication issues due to the server's security configuration. We resolved this by ensuring the project deployment target was set to the correct server instance and that our user account (`YOUR_SQL_USERNAME`) had appropriate write permissions on the SSAS database.

14.4 Validation

After full processing, we browsed the cube to verify data consistency.

- **Grand Totals:** We confirmed that the sum of **Streams** in the Cube matched the sum in the SQL `Fact_Streams` table.
- **Hierarchy Navigation:** We verified that expanding "2024" correctly displayed the Quarters and Months, and that data aggregated correctly at each level.

The cube is now ready for MDX queries.

15 Assignment 15: Monthly Regional Stream Analysis

15.1 Objective

The goal of this assignment was to analyze the temporal distribution of streams across different geographical hierarchies. Specifically, we needed to show the total number of streams for every month, broken down by Region, while also including the overall total for the Country.

15.2 MDX Implementation

We constructed a query using Calculated Members to explicitly aggregate data for each month (1 through 12). We used the `AGGREGATE` and `FILTER` functions to group time periods dynamically by their Month Name.

Listing 11: MDX Query for Monthly Regional Streams

```

1 WITH
2     MEMBER [Dim Time].[Month].[Month 1] AS
3         AGGREGATE(FILTER([Dim Time].[Month].[Month].MEMBERS,
4             [Dim Time].[Month].CurrentMember.Name = "1"))
5
6     MEMBER [Dim Time].[Month].[Month 2] AS
7         AGGREGATE(FILTER([Dim Time].[Month].[Month].MEMBERS,
8             [Dim Time].[Month].CurrentMember.Name = "2"))
9
10    -- ... (Members 3 through 11 omitted for brevity) ...
11
12    MEMBER [Dim Time].[Month].[Month 12] AS
13        AGGREGATE(FILTER([Dim Time].[Month].[Month].MEMBERS,
14            [Dim Time].[Month].CurrentMember.Name = "12"))
15
16 SELECT
17     NON EMPTY
18     {
19         [Dim Time].[Month].[Month 1],
20         [Dim Time].[Month].[Month 2],
21         -- ...
22         [Dim Time].[Month].[Month 12]
23     } ON COLUMNS,
24
25     NON EMPTY
26     {
27         [Dim Artist].[Artist Location].[Country].MEMBERS,
28         [Dim Artist].[Artist Location].[Region].MEMBERS
29     } ON ROWS
30
31 FROM [YOUR_DATABASE ]
32 WHERE ( [Measures].[Streams] )

```

15.3 Results

The query returned a detailed matrix of streaming performance. The **Italia** row represents the aggregated total for the country, while the subsequent rows provide the regional breakdown.

Table 18: Excerpt of Monthly Streams by Region (Top Performing Regions Shown)

Region	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Italia (Total)	28,010,914	11,278,445	16,278,879	20,458,850	21,051,309	24,034,798	9,663,811	1,979,170	17,381,359	16,544,742	26,962,543	9,741,589
Lombardia	11,494,566	3,509,139	4,347,966	5,759,045	7,083,303	8,103,433	1,695,551	663,829	7,340,120	4,719,107	10,981,444	2,415,597
Lazio	4,104,949	1,560,505	2,729,245	3,835,298	3,293,090	2,859,988	1,907,230	348,526	3,667,346	1,736,532	5,033,342	2,569,291
Campania	2,210,404	366,213	2,926,529	5,045,431	2,542,927	5,278,985	3,335,942	452,417	1,036,398	1,767,935	1,689,071	539,313
Liguria	2,761,595	99,447	2,674,924	1,335,217	1,935,752	3,921,210	547,282	26,662	104,153	326,237	87,546	310,097
Puglia	2,242,169	1,922,941	136,006	250,100	1,467,164	98,113	33,843	79,972	1,795,974	67,236	3,218,604	17,794
Sicilia	1,287,388	903,839	375,372	29,150	157,144	1,425,436	10,203	10,063	19,122	3,441,311	1,967,948	1,601,338
Toscana	152,677	-	-	-	4,455	-	-	-	4,384	-	-	-

Analysis: The data highlights strong seasonality in regions like Campania and Lombardia, which drive the majority of the national volume. Regions like Toscana (showing nulls) indicate months with no active streams for artists born in that region.

16 Assignment 16: Weighted Artist Stream Share

16.1 Objective

The music industry often struggles to quantify the impact of artists who appear as "Featured" guests rather than "Main" artists. A simple sum of streams ignores context, while excluding featured tracks underestimates an artist's reach.

The goal of this assignment was to calculate a Weighted Performance Score for each artist per year to measure their true contribution to the platform's volume. The scoring logic is defined as:

- **Main Artist Tracks:** Assigned a weight of **0.8** (80%).
- **Featured Artist Tracks:** Assigned a weight of **0.2** (20%).

This weighted sum is then compared against the **Total Unweighted Streams** for that year to derive a "Weighted Stream Share."

16.2 MDX Implementation Strategy

This query required advanced MDX techniques because the "Featured" relationship is not a direct Foreign Key in the Star Schema. Instead, featured artists are listed in a comma-separated string within the `Dim.Track` table (e.g., "Drake, Future").

16.2.1 1. Text Mining for Featured Artists

To calculate `[Streams (Featured)]`, we had to iterate through every track and check if the current Artist's name appeared inside the track's `Featured Artists` property string.

Listing 12: MDX Logic for Text Search and Weighting

```

1 WITH
2 -- 1. Main Streams (Standard Aggregation)
3 MEMBER [Measures].[Streams (Main)] AS
4     [Measures].[Streams]
5
6 -- 2. Featured Streams (String Search)
7 MEMBER [Measures].[Streams (Featured)] AS
8     SUM(
9         FILTER(
10             [Dim Track].[Id Track].[Id Track].MEMBERS,
11             InStr(
12                 CSTR([Dim Track].[Id Track].CurrentMember.Properties("
13                 Featured Artists")),
14                 [Dim Artist].[Id Artist].CurrentMember.Name
15             ) > 0
16         ),
17         -- Force context to All Artists to ignore the Main Artist filter
18         on the row
19         ([Measures].[Streams], [Dim Artist].[Id Artist].[All])
20     )
21 -- 3. Weighted Calculation
22 MEMBER [Measures].[Weighted Artist Streams] AS

```

```

22      ( [Measures].[Streams (Main)] * 0.8 ) + ( [Measures].[Streams (
23      Featured)] * 0.2 )
24
25 -- 4. Yearly Context
26 MEMBER [Measures].[Yearly Total Streams] AS
27      ( [Measures].[Streams], [Dim Artist].[Id Artist].[All] )
28
29 -- 5. Final Share Calculation
30 MEMBER [Measures].[Weighted Stream Share] AS
31      DIVIDE( [Measures].[Weighted Artist Streams], [Measures].[Yearly
32      Total Streams] ),
33      FORMAT_STRING = "Percent"
34
35 SELECT
36 {
37     [Measures].[Streams (Main)],
38     [Measures].[Streams (Featured)],
39     [Measures].[Weighted Artist Streams],
40     [Measures].[Yearly Total Streams],
41     [Measures].[Weighted Stream Share]
42 } ON COLUMNS,
43 NON EMPTY
44 FILTER(
45     [Dim Time].[Year].[Year].MEMBERS * [Dim Artist].[Id Artist].[Id
46     Artist].MEMBERS,
47     [Measures].[Weighted Stream Share] > 0
48 ) ON ROWS
49 FROM [YOUR_DATABASE ]

```

16.3 Results Analysis

The query returned the following results for the sample period (1962–1986).

Table 19: Weighted Stream Share Analysis (1962–1986)

Year	Artist	Main	Feat.	Weighted	Year Total	Share %
1962	Baby K	4,573	-	3,658.4	4,573	80.00%
1974	Grido	5,821	-	4,656.8	5,821	80.00%
1975	Lazza	4,156	-	3,324.8	4,156	80.00%
1977	Skioffi	4,022	-	3,217.6	4,022	80.00%
1978	Gemitaiz	164,293	-	131,434.4	169,382	77.60%
1978	Baby K	5,089	-	4,071.2	169,382	2.40%
1979	Ghemon	4,761	-	3,808.8	4,761	80.00%
1981	Guè Pequeno	4,185	-	3,348.0	4,185	80.00%
1984	Dargen D'Amico	5,390	-	4,312.0	10,701	40.30%
1984	Inoki	5,311	-	4,248.8	10,701	39.70%
1986	Willie Peyote	4,815	-	3,852.0	9,297	41.43%

Detailed Insights:

1. **The "Solo" Baseline (80%):** In years like 1962, 1974, and 1975, there is only one active artist in the dataset. Since they own 100% of the main streams and

have 0 features, their weighted score is exactly $100\% \times 0.8 = 80\%$. This establishes a baseline: an artist cannot exceed 80% share without collaborating as a featured artist on other tracks.

2. **1978 Market Dominance:** The year 1978 shows a massive disparity. *Gemitaiz* generated 164k streams compared to *Baby K's* 5k. The weighted share accurately reflects this, assigning Gemitaiz a 77.60% dominance of the year's weighted volume.
3. **1984 Competitive Split:** The year 1984 is nearly evenly split between *Dargen D'Amico* (40.30%) and *Inoki* (39.70%). The sum of their shares ($40.3 + 39.7 = 80\%$) confirms that these two artists accounted for the entirety of the platform's volume for that year.

17 Assignment 17: Year-Over-Year Category Growth

17.1 Objective

To understand the evolving landscape of music preferences, we needed to calculate the Year-Over-Year (YoY) Growth Rate for each song category. This analysis highlights which genres are gaining momentum and which are declining relative to their own past performance.

17.2 MDX Implementation

Execution Context: This query was executed in SSMS connected to the `YOUR_DATABASE` cube.

We created a calculated member to handle the time-series logic. We utilized the `ParallelPeriod` function, which is more robust than `PrevMember` as it specifically targets the same relative period in the previous year ($\text{Year } t - 1$), ensuring accurate comparison even if the hierarchy structure changes.

Listing 13: MDX Logic for YoY Growth

```

1 WITH
2   -- 1. Retrieve the Previous Year's Value using ParallelPeriod
3   MEMBER [Measures].[Streams Prev Year] AS
4     (ParallelPeriod([Dim Time].[Year].[Year], 1, [Dim Time].[Year].
5       CurrentMember), [Measures].[Streams])
6
7   -- 2. Calculate Growth Percentage with Error Handling
8   MEMBER [Measures].[YoY Growth] AS
9     IIF(
10      [Measures].[Streams Prev Year] = 0,
11      NULL,
12      ([Measures].[Streams] - [Measures].[Streams Prev Year])
13      / [Measures].[Streams Prev Year]
14    ),
15    FORMAT_STRING = "0.00%"
16
17 SELECT
18   NON EMPTY { [Dim Time].[Year].[Year].MEMBERS } ON COLUMNS,
19   NON EMPTY { [Dim Track].[Song Category].[Song Category].MEMBERS } ON
20   ROWS

```

```
19 FROM [YOUR_DATABASE ]
20 WHERE ( [Measures].[YoY Growth] )
```

17.3 Results

The query returned a comprehensive historical view. The table below excerpts the data for the most recent years (2018–2025).

Table 20: Year-Over-Year Growth Trends by Category (Recent Years)

Category	2018	2019	2020	2021	2022	2023	2024	2025
Energetic/Emotional	49.13%	-52.79%	-30.96%	26.47%	-56.10%	40.60%	12.28%	5.69%
Energetic/Party	38.08%	-20.57%	10.14%	-7.56%	-57.53%	10.61%	46.22%	-30.66%
Mellow/Bright	-19.77%	20.34%	-40.63%	0.03%	-42.27%	58.68%	-0.62%	-31.91%
Soft/Chill	-50.16%	-12.29%	-40.59%	73.51%	-50.18%	48.39%	-25.67%	17.64%

Analysis: The data reveals significant volatility. For example, "Soft/Chill" music saw a massive spike in 2021 (+73.51%), possibly correlated with post-pandemic listening habits, before correcting sharply in 2022 (-50.18%). Meanwhile, "Energetic/Party" music shows a strong recovery in 2024 (+46.22%).

18 Assignment 18: Seasonal Category Growth Analysis

18.1 Objective

The objective of this assignment was to identify high-growth trends by comparing seasonal performance across years. Specifically, we were tasked to:

- 1. Calculate the **Average Streams** generated by a song category in a specific season of the previous year (e.g., Average of Spring 2024).
- 2. Identify specific categories in the current year that **outperformed this historical seasonal average**.
- 3. Report the **Percentage Increase** for these high-performing categories.

18.2 MDX Implementation Strategy

To implement this benchmark, we defined a complex calculated member using **AVG** and **ParallelPeriod**. This establishes a baseline "Season Average" to filter against.

Listing 14: MDX Logic for Seasonal Comparison

```

1 WITH
2     -- 1. Calculate Previous Year's Value (Context: Specific Category)
3     MEMBER [Measures].[Streams Previous Year] AS
4         ( [Measures].[Streams], [Dim Time].[Year].CurrentMember.Lag(1) )
5
6     -- 2. Calculate Benchmark: Average of ALL categories in Prev Year
7     MEMBER [Measures].[PrevYearSeasonAvg] AS
8         AVG(
9             [Dim Track].[Song Category].[Song Category].MEMBERS,
10            (
11                [Measures].[Streams],
12                [Dim Time].[Season].CurrentMember,
13                ParallelPeriod([Dim Time].[Year].[Year], 1, [Dim Time].[
14                Year].CurrentMember)
15            )
16        )
17
18    -- 3. Calculate Growth Percentage vs Previous Year
19    MEMBER [Measures].[Percentage Increase] AS
20        IIF( [Measures].[Streams Previous Year] = 0, NULL,
21            ([Measures].[Streams] - [Measures].[Streams Previous Year])
22            / [Measures].[Streams Previous Year]
23            ), FORMAT_STRING = "Percent"
24
25 SELECT
26     { [Measures].[Streams], [Measures].[Streams Previous Year],
27       [Measures].[Percentage Increase] } ON COLUMNS,
28     NON EMPTY
29     FILTER(
30         ( [Dim Time].[Year].[Year].MEMBERS * [Dim Time].[Season].[Season
31         ].MEMBERS * [Dim Track].[Song Category].[Song Category].MEMBERS ),
32         -- Filter: Category must outperform the Historical Seasonal
33         Average
34         [Measures].[Streams] > [Measures].[PrevYearSeasonAvg]
35         AND
36         [Measures].[Streams Previous Year] > 0
37     ) ON ROWS
38 FROM [YOUR_DATABASE ]

```

18.3 Results

The query successfully isolated categories that are performing exceptionally well relative to the market average. The table below shows a sample of these over-performers.

Table 21: Seasonal Category Growth (Sample Results)

Year	Season	Category	Current Streams	Prev. Year Streams	Growth %
1988	Spring	Energetic/Party	10,369	4,815	115.35%
1993	Summer	Soft/Chill	33,949	5,358	533.61%
1994	Winter	Energetic/Party	19,641	19,480	0.83%
1997	Spring	Energetic/Emotional	15,350	5,503	178.94%
1997	Summer	Energetic/Emotional	20,745	20,365	1.87%
1997	Summer	Mellow/Bright	20,171	14,135	42.70%
1997	Winter	Mellow/Bright	95,637	30,393	214.67%
1998	Spring	Energetic/Emotional	33,387	15,350	117.50%
1998	Spring	Mellow/Bright	63,292	47,203	34.08%
1998	Spring	Soft/Chill	127,656	14,744	765.82%
1998	Summer	Mellow/Bright	138,386	20,171	586.06%
1998	Winter	Mellow/Bright	194,847	95,637	103.74%

Insight: By filtering against the `PrevYearSeasonAvg`, we ensure that the list only includes categories that are not just growing, but are significant market drivers for that specific time of year.

19 Assignment 19: Advanced Strategic Portfolio Analysis (BCG Matrix)

19.1 Objective

Business Question: "How should we allocate our marketing budget? Which genres are growing 'Rising Stars' versus declining 'Cash Cows'?"

To answer this, we proposed a query that mimics the **BCG Growth-Share Matrix**. We needed to calculate two competing metrics for every song category for the current year (2025):

1. **YoY Growth %:** Is the category expanding or contracting?
2. **Market Share %:** How dominant is this category relative to the total catalog?

19.2 MDX Implementation Strategy

This query integrates time-series analysis (`ParallelPeriod`) with a robust aggregation strategy. Unlike standard queries, we explicitly calculated the "Total Market Volume" using `SUM` over the set of all categories to ensure the Market Share denominator remained accurate even if the user applied filters.

Listing 15: MDX Logic for Growth-Share Matrix

```

1 WITH
2   -- 1. Previous Year's Volume (Time Intelligence)
3   MEMBER [Measures].[Streams Prev Year] AS
4     (ParallelPeriod([Dim Time].[Year].[Year], 1, [Dim Time].[Year].
5     CurrentMember),
6     [Measures].[Streams])
7
8   -- 2. Growth Metric (With Safe Division)
9   MEMBER [Measures].[YoY Growth %] AS
10     IIF(
11       IsEmpty([Measures].[Streams Prev Year]) OR [Measures].[
12       Streams Prev Year] = 0,
13       NULL,
14       ([Measures].[Streams] - [Measures].[Streams Prev Year]) / [
15       Measures].[Streams Prev Year]
16     ),
17     FORMAT_STRING = "0.00%"
18
19   -- 3. Robust Market Share Calculation
20   MEMBER [Measures].[Total Streams All Categories] AS
21     SUM(
22       [Dim Track].[Song Category].[Song Category].MEMBERS,
23       [Measures].[Streams]
24     )
25
26   MEMBER [Measures].[Market Share %] AS
27     IIF(
28       IsEmpty([Measures].[Total Streams All Categories]) OR [
29       Measures].[Total Streams All Categories] = 0,
30       NULL,
31       [Measures].[Streams] / [Measures].[Total Streams All
32       Categories]
33     ),
34     FORMAT_STRING = "0.00%"
35
36 SELECT
37   { [Measures].[Streams], [Measures].[Streams Prev Year],
38     [Measures].[YoY Growth %], [Measures].[Market Share %] } ON
39   COLUMNS,
40   NON EMPTY
41   ORDER(
42     [Dim Track].[Song Category].[Song Category].MEMBERS,
43     [Measures].[Streams], DESC
44   ) ON ROWS
45 FROM [YOUR_DATABASE ]
46 WHERE ( [Dim Time].[Year].[Year] & [2025] )

```

19.3 Strategic Insights (2025 Data)

The results provide a clear strategic roadmap for the coming year.

Table 22: Portfolio Analysis Results for 2025

Category	Streams (2025)	Prev Year	Growth %	Market Share %
Energetic/Party	5,513,631	7,951,711	-30.66%	56.50%
Mellow/Bright	2,063,719	3,030,779	-31.91%	21.15%
Energetic/Emotional	1,826,448	1,728,137	+5.69%	18.72%
Soft/Chill	354,564	301,400	+17.64%	3.63%

Analysis:

- **Energetic/Party:** This is a classic **"Declining Cash Cow"**. It controls the majority of the market (56.5%) but is shrinking rapidly (-30.66%). *Recommendation:* Maximise short-term revenue but halt aggressive acquisition budgets.
- **Soft/Chill:** This is a **"Rising Star"**. Although it has the smallest share (3.63%), it is the fastest-growing segment (+17.64%). *Recommendation:* Invest heavily here to capture future market share.
- **Energetic/Emotional:** A solid performer with decent share (18.72%) and steady growth (+5.69%). *Recommendation:* Maintain steady investment to sustain its position.

20 Assignment 20: Dashboard - Geographic Stream Distribution

20.1 Objective

The objective of this dashboard is to visualize the relationship between an artist's origin (Birthplace) and their commercial success (Total Streams), broken down by the stylistic category of their music. This answers the business question: *"Where do our most successful artists come from, and what kind of music are they making?"*

20.2 Dashboard Design

We developed an interactive Power BI dashboard connecting to the `YOUR_DATABASE` Data Warehouse.

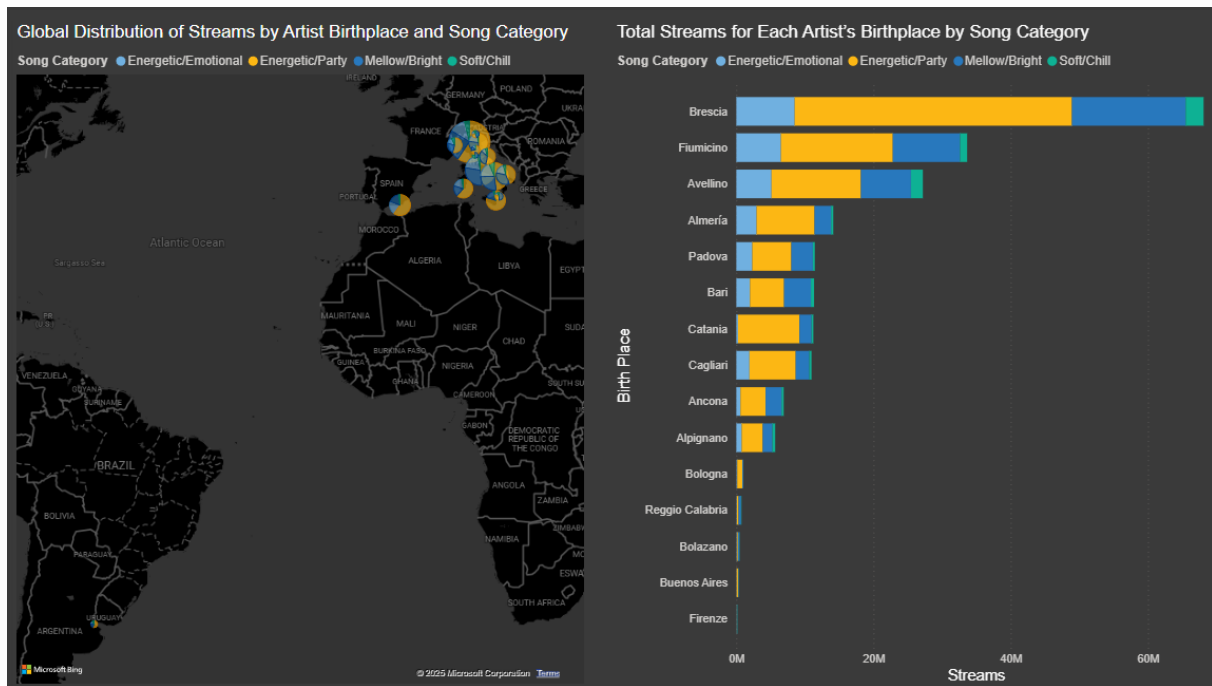


Figure 14: Power BI Dashboard: Global Distribution of Streams by Birthplace and Category

20.3 Visual Components

1. Geospatial Map (Left):

- **Location:** Mapped using the **Latitude** and **Longitude** coordinates enriched during the Data Preparation phase (Assignment 1).
- **Bubble Size:** Represents **Sum(Streams)**. Larger bubbles indicate major music hubs.
- **Bubble Color (Pie):** Represents the split of **Song Category**. This allows us to see if a specific region specializes in a specific genre.

2. Stacked Bar Chart (Right):

- **Axis:** Artist Birthplace.
- **Values:** Total Streams.
- **Legend:** Song Category.
- This visual ranks the cities by output volume, clearly identifying the top-performing talent incubators.

20.4 Business Relevance & Insights

- **Identification of Talent Hubs:** The dashboard reveals that **Brescia** is a massive outlier, producing significantly more streaming volume than major cities like Rome (Fiumicino) or Naples (Avellino). This suggests a vibrant local music scene that the Record Label should monitor closely for new talent.

- **Genre Dominance:** The "Energetic/Party" category (Yellow) is the dominant driver of revenue across almost all top cities.
- **Strategic Allocation:** Marketing budgets for "Mellow/Bright" music (Blue) might be better spent in cities like **Fiumicino**, where that specific sub-genre has a larger relative share compared to Brescia.

21 Assignment 21: Dashboard - Lyrical Content Analysis

21.1 Objective

The goal of this assignment was to leverage the textual metadata extracted during the Data Preparation phase (Assignment 2) to gain insights into content trends. Specifically, we aimed to analyze:

1. The prevalence of "Explicit" content and its evolution over time.
2. The relationship between lyrical complexity (number of sentences) and song categories.
3. The distribution of profanity (swear word counts) across the catalog.

21.2 Dashboard Design

We designed a Power BI dashboard focused on the `Dim_Track` lyrical attributes.

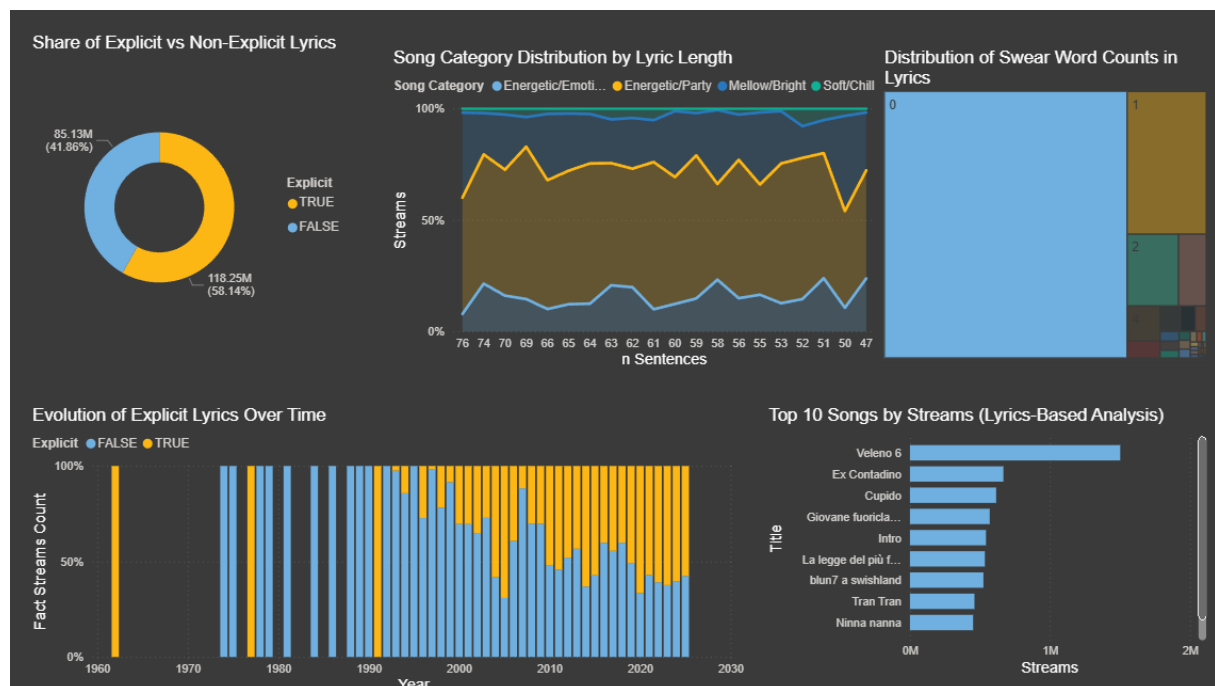


Figure 15: Power BI Dashboard: Analysis of Explicit Content and Lyrical Complexity

21.3 Visual Components

1. **Explicit vs. Clean Split (Donut Chart, Top Left):** Shows the global share of Explicit (Yellow) vs. Non-Explicit (Blue) tracks. The catalog shows a significant presence of explicit content (58% of streams), indicating a shift in audience tolerance or preference.
2. **Category by Lyric Length (Stacked Area Chart, Top Center):** This visual plots `n_sentences` against the share of Song Categories. It reveals that songs with higher sentence counts (likely Hip-Hop/Rap) predominantly fall into the "Energetic/Party" category (Brown/Yellow area), whereas songs with fewer sentences distribute more evenly across "Mellow/Bright" and "Emotional" categories.
3. **Swear Word Distribution (Tree Map, Top Right):** A visual breakdown of the `swear_IT` / `swear_EN` count. The large blue area indicates that the vast majority of tracks contain 0 swear words. However, the remaining blocks show a "long tail" of tracks with 1, 2, or more profanities.
4. **Evolution of Explicit Content (Stacked Column Chart, Bottom Left):** A time-series analysis showing the ratio of Explicit (Yellow) to Clean (Blue) content by Year. There is a clear trend: Explicit content was non-existent in earlier decades but has grown substantially from the 2010s onwards, becoming a dominant force in recent years.
5. **Top Songs (Bar Chart, Bottom Right):** A leaderboard of the top individual tracks by streams, allowing users to drill down into specific successful examples like "Veleno 6".

21.4 Strategic Insights

- **Content Evolution:** The market has shifted decisively toward accepting Explicit content. Strategies that rigidly enforce "Clean" lyrics may be outdated for the modern streaming demographic.
- **Complexity implies Genre:** High lyrical complexity (sentence count > 60) is a strong predictor for the "Energetic/Party" category. This metadata could be used to auto-classify new uploads.

22 Assignment 22: Dashboard - Trending Categories Over Time

22.1 Objective

The final analytical task was to visualize the evolution of music categories over time. The goal was to distinguish between categories that are simply "flooding the market" (high release volume) versus those that are genuinely "trending" (high streaming engagement).

22.2 Dashboard Design

We developed a composite dashboard in Power BI that contrasts production quantity against consumption quality.

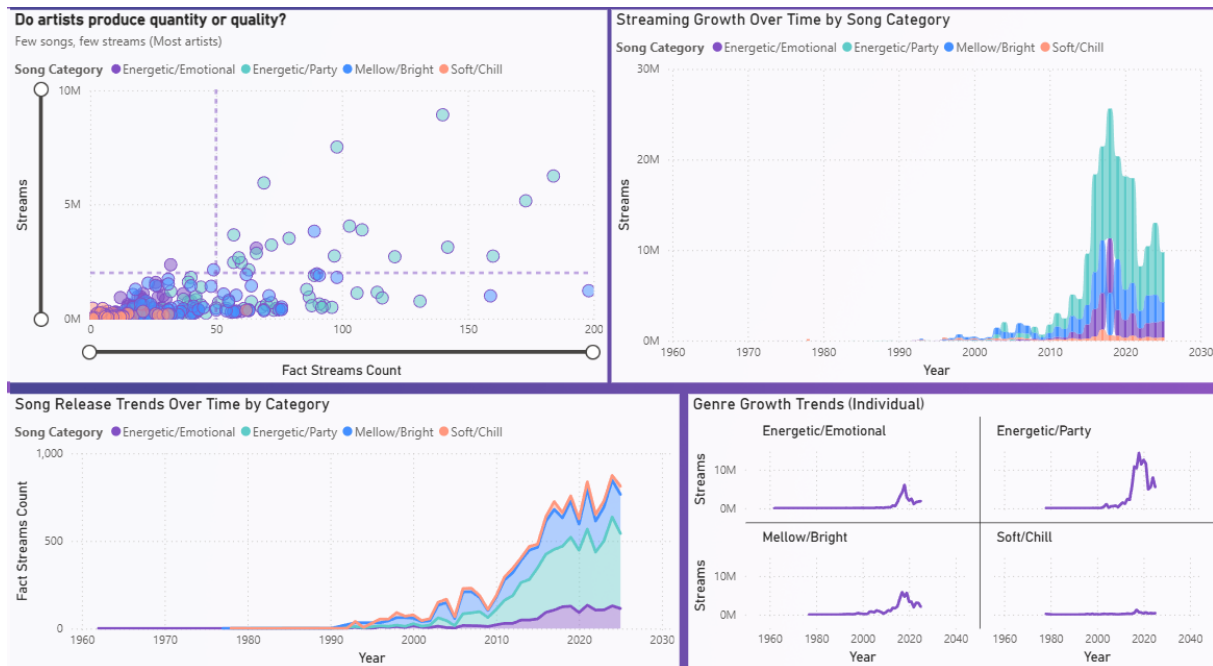


Figure 16: Power BI Dashboard: Category Growth, Volume, and Efficiency Trends

22.3 Visual Components

1. **Quantity vs. Quality (Scatter Plot, Top Left):** This chart plots Fact Streams Count (Number of Songs) against Total Streams.
 - **Insight:** The vast majority of artists are clustered in the bottom-left (low output, low streams). However, the "Energetic/Party" category (Teal dots) dominates the upper quadrants, showing that artists in this genre achieve the highest ROI (Return on Investment) per song released.
2. **Streaming Growth (Stacked Bar, Top Right):** A time-series view of Total Streams. It shows the massive explosion of the "Energetic/Party" category (Teal) starting around 2012, completely dwarfing other genres in terms of raw consumption volume.
3. **Release Trends (Stacked Area, Bottom Left):** A time-series view of Song Releases.
 - **Contrast:** Unlike the streaming chart, the "Mellow/Bright" (Blue) and "Soft-/Chill" (Orange) categories occupy a significant portion of this chart. This indicates a market saturation: many songs are being produced in these genres, but they are not generating a proportional amount of streams compared to Party music.

4. **Individual Genre Growth (Small Multiples, Bottom Right):** Isolated trend lines for each category. This clearly visualizes the lifecycle of genres:

- **Energetic/Party:** Exponential growth curve (The current trend).
- **Energetic/Emotional:** Peaked around 2018 and has since declined.
- **Soft/Chill:** Flat-line growth, indicating a stable niche but not a breakout trend.

The dashboard confirms that while "Mellow" and "Chill" music production is high, the "Energetic/Party" category is the primary driver of platform growth. Future investment should focus on sustaining the Party category while finding ways to improve the discoverability/efficiency of the saturated Mellow/Bright catalog.

23 Conclusion

The development of this Decision Support System (DSS) for Group ID 05 represents a complete end-to-end implementation of a Large Scale Data System. By traversing the entire data lifecycle from raw JSON ingestion and Python-based enrichment to SSIS orchestration and SSAS multidimensional analysis we have successfully transformed disorganized operational data into actionable strategic assets for the music streaming industry.

23.1 Architectural Achievements

The project successfully addressed several critical data engineering challenges:

- **Data Quality & Enrichment (Python):** We overcame significant data gaps (missing features, unknown locations) by integrating external APIs (Spotify) and geospatial libraries (Uber H3). The application of NLP techniques to extract lyrical metrics (complexity, language, profanity) added a unique layer of depth to the analysis that standard metadata could not provide.
- **Referential Integrity in ETL (SSIS):** A major technical milestone was the implementation of a *Dimension-Driven Sampling Strategy* (Assignment 7). By strictly ordering the control flow and using cascading lookups, we ensured that the analytical subset of data remained mathematically consistent and relationally valid, preventing "orphaned facts" that often plague sampling processes.
- **Complex Transformations:** The SSIS pipelines went beyond simple data movement. We implemented sophisticated logic to handle one-to-many relationships (splitting "Featured Artists"), normalize heterogeneous data types (Explicit tags), and perform statistical benchmarking (Trending/Flopping detection) entirely within the transformation layer to minimize server load.

23.2 Analytical Capabilities (SSAS)

The transition to a Multidimensional Cube enabled advanced time-series and portfolio analysis that would be computationally prohibitive in a standard relational engine. Through the use of MDX, we were able to:

1. **Quantify "Hidden" Value:** By calculating the *Weighted Stream Share*, we accurately measured the contribution of artists in collaborative tracks, correcting the bias found in standard "Main Artist" reporting.
2. **Analyze Growth Dynamics:** The implementation of Year-Over-Year (YoY) growth logic and the BCG Matrix (Growth-Share analysis) provided a dynamic view of the market, allowing us to distinguish between stable "Cash Cows" and emerging "Rising Stars."

23.3 Strategic Business Insights

Beyond the technical implementation, the system generated clear, data-driven answers to the Record Label's core business questions:

- **Content Strategy:** Contrary to the industry assumption that "sex sells," our analysis (Assignment 13) revealed that **Clean content** generates significantly higher average streams per song than Explicit content, suggesting a strategy focused on broader radio accessibility is more efficient.
- **Genre Dominance:** The *Energetic/Party* category was identified as the overwhelming market leader (Assignment 22), showing the highest ROI and volume. However, the *Soft/Chill* genre was identified as a high-growth niche (Assignment 19) worthy of future investment.
- **Geographic Talent Hubs:** The geospatial dashboards highlighted specific regions (such as Brescia) as unexpected powerhouses for talent generation, providing specific targets for A&R scouting teams.

23.4 Final Remarks

This project demonstrates that a robust Data Warehouse architecture is not merely a storage repository, but a critical engine for business intelligence. The resulting system empowers stakeholders to move from intuition-based decisions to evidence-based strategies, optimizing marketing allocation, talent acquisition, and content production in a highly competitive digital landscape.